

Dealerless GPU Preprocessing for Orca Linear Layers

Internal/Advisor Technical Checkpoint

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August 17, 2026

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Abstract

We implement a two-process GPU preprocessing path that replaces Orca’s trusted dealer for forward linear layers at unpinned Ring-LPN feasibility parameters. Party-local sparse noise feeds distributed DPF generation, Ring-LPN OLE, noncircular Ring-OLE-output bootstrap, exact $\mathbb{Z}_Q \rightarrow \mathbb{Z}_{2^{bw}}$ conversion, stock-format key serialization, and unchanged Orca FC/Conv2D consumers. The default SCI/IKNP focused suite and a separate opt-in EMP-Silent rerun each pass the same five q64/q128 cases and sixteen rejection controls; the latter is independently unreviewed correctness/accounting evidence only. The controlled source-manifest matrix covers both CNN2 FC layers and the CNN3 FC scale point. Every layer passes one warmup plus ten measured trials. On a quiescent RTX 5000 Ada selected per sample for both the protocol critical path and subsequent stock dealer, ratios of summed per-layer mean setup-included preprocessing and stock-dealer times are $882.33\times$ for CNN2’s two FC layers and $60.58\times$ for CNN3 FC5. These same-host feasibility results are a controlled negative comparison, not a speedup or security claim. A source-native terminal FC/Conv2D application also passes nonzero input-share and public-bias checks; the extracted stock helpers pass signed/wrapping clear oracles. Arbitrary multi-layer mask-state chaining remains open.

A source-bound known-zero control separately composes all 21 fresh linear records through the exact 62-item ResNet18 stream. Its trusted test-only adapter reads both parties’ mask states, supplies both truncation successor-mask shares and remask/terminal material, and generates the nonlinear keys. This proves graph/state compatibility only—not dealerless nonlinear preprocessing, private/trained inference, or full-model performance. Mutual HMAC-SHA256 authenticates endpoint roles and invocation context before local preflight, but later protocol bytes have no per-message integrity. The exercised source tuple $(n, c, t) = (8192, 2, 8)$ and full-graph execution override $(262144, 2, 8)$ are correctness points, not concrete-security pins. Authenticated distinct-host execution, a compatible dealerless baseline, qualified independent review, clean-clone evidence, and final release identities remain open. Accordingly, this document is an internal technical checkpoint, not a publication-ready security or systems result.

Artifact and claim discipline. Historical numerical claims use the retained manifests/CSVs named in §5.3; the September review experiment is separately identified in §6. Historical source, binary, schema, and evidence digests are checked by the release gate. q64/q128 denote one/two approximately 62-bit CRT limbs, not security levels. “Authenticated” for the current same-host measurements means only the preflight endpoint/context handshake; the conditional theorem still assumes authenticated channels.

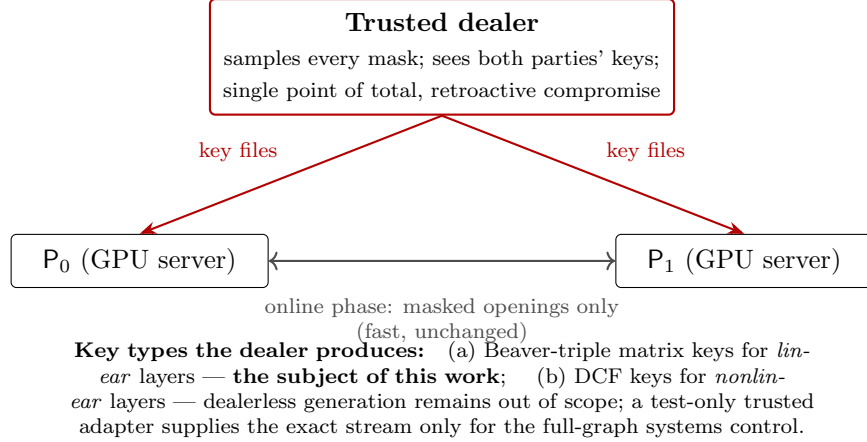


Figure 1: Orca’s preprocessing today. The dealer is trusted with every secret of the offline phase. This proposal removes it for the linear-layer keys; the online phase is never modified.

1 Executive status: what is implemented, what is not

Thesis. This work has two deliberately separate results: (i) an executable forward-linear preprocessing pipeline without a trusted dealer in the stated random-oracle model, validated by unchanged Orca consumers; and (ii) a known-zero full-graph compatibility control whose nonlinear/truncation state is explicitly trusted. The system result is the composition, GPU/system design, fail-closed state machinery, and evidence boundary; Ring-LPN, DPF, OT/OLE, conversion, Beaver multiplication, and Orca remain prior components. No concrete Ring-LPN parameter or compatible dealerless-PCG performance comparison is established.

2 Problem overview: Orca’s preprocessing and its trusted dealer

2.1 Background: Orca in one page

Orca [1] is a two-party secure computation (2PC) system for neural-network training and inference, engineered around GPUs. Two servers P_0 and P_1 hold additive secret shares of all inputs, weights, and activations; neither learns anything beyond its own shares. Like most practical 2PC systems, Orca splits work into an input-independent *preprocessing* (offline) phase that generates correlated randomness, and a fast *online* phase that consumes it:

- **Linear layers** (convolutions, fully-connected/FC layers) consume *Beaver-triple-style matrix keys*: for a layer multiplying an $M \times K$ matrix by a $K \times N$ matrix, the parties need shares of random masks A, B and of the product terms that make the masked multiplication identity work. Online, each party opens masked operands and runs the local computation implemented by `gpuMatmulBeaver`.
- **Nonlinear layers** (ReLU, truncation) consume function secret sharing (FSS) keys — distributed comparison functions (DCF) — with a different structure.

In stock Orca *both* kinds of keys are produced by a **trusted dealer**: a third machine that samples every mask, computes every correlated value, and ships each party its key files (Figure 1).

2.2 Current status of our pipeline

Over the preceding milestones this project built, and validated on GPUs, a preprocessing pipeline that produces Orca’s FC-layer keys from *Ring-LPN pseudorandom correlation generators* rather than from a dealer’s coin flips. The pipeline (Figure 2; components defined in §4) already works end-to-end with *real* correlation generators — not idealized stand-ins — at both supported modulus widths, and its output keys are validated by the strongest referee available: Orca’s *unchanged* online code reconstructs the correct products (§5.3 gives the measured evidence).

2.3 Historical gaps and current boundary

The original one-process transcript had four dealer/security gaps. The live forward artifact closes the first three at the executable level; the fourth still blocks a deployable security claim. Dealerless nonlinear preprocessing remains outside the theorem; a narrow trusted-adaptor systems control now executes the stock nonlinear consumers.

- G1 — Centralized SPFSS setup: closed in the live path.** Each process samples only its own sparse noise, participates in distributed GPU-AES DPF key generation, and expands only its own Ring-LPN share. The older one-process transcript retains `build_spfss_keys()` solely as a diagnostic/validation artifact; none of its timings are presented as live protocol measurements.
- G2 — Exact share conversion: implemented; privacy proof reopened.** Batched `edaBits`, `daBits`, and Boolean triples from the selected OT backend implement exact $\mathbb{Z}_Q \rightarrow \mathbb{Z}_{2^{bw}}$ conversion. The canonical and classifier suites use SCI/IKNP. A current separate EMP-Silent rerun passes the same five focused cases and sixteen controls with exact directional inventory exhaustion, but remains opt-in and independently unreviewed. The wrap bit is never opened directly. Section 4.5 corrects the local output-conditioned inverse, but a full state-consistent conditional transcript proof remains open (P-CONV). The measured local runner first performs mutual HMAC-SHA256 endpoint/context establishment; later loopback bytes have no application-layer per-message integrity.
- G3 — Randomness/RO separation: closed.** Private roots, masks, and noise come from each process’s OpenSSL private DRBG. For every Ring-OLE correlation, $a_0 = 1$ is the unsent public identity polynomial. Each party exchanges four uniform 64-bit seed-share words; their XOR is one jointly sampled 256-bit public seed. The full 256-bit correlation scope and all public Ring-OLE parameters domain-separate SHAKE256 rejection sampling of every $(c-1)n$ -coefficient tail, giving independent exact-uniform ring elements in the explicit random-oracle model rather than an information-theoretic coin-toss claim.
- G4 — Security parameters: open.** The deployed $(c, t) = (2, 8)$ values are feasibility parameters. The exact splittable Ring-LPN distribution and the two-CRT-limb composition have no reviewed concrete classical/quantum advantage bound or accepted parameter pin.
- G5 — Scope.** Stateful training and dealerless nonlinear DCF key generation are not implemented. The proved/exercised security scope remains forward FC. A separate source-bound known-zero systems control executes the complete stock ResNet18 graph with a trusted nonlinear key source; “dealerless Orca” without the forward-linear qualifier remains false.

Contributions. (1) A live, trusted-dealer-free two-process forward-FC/Conv2D preprocessing path in the stated random-oracle model, whose party-local outputs are consumed by unchanged Orca code; (2) a source-specific exact algebraic coupling of distributed DPF correction words to standard generation, complete role-specific batch simulators, and an executable compatibility repair that bootstraps every post-base Phase-C product from Ring-OLE output; (3) a tree-block GPU Phase-C implementation whose correctness and component timings are retained without claiming an end-to-end speedup; (4) a hybrid-model privacy proof and live realization of exact $\mathbb{Z}_Q \rightarrow \mathbb{Z}_{2^{bw}}$ conversion; (5) a source-to-transcript map and conditional forward-FC theorem with explicit leakage, assumptions, and nonclaims; (6) a source-manifest-derived CNN2/CNN3 forward-FC feasibility matrix with one warmup and ten passing trials per layer and a same-physical-GPU, quiescence-checked stock trusted-dealer comparator; (7) a fail-closed source-bound known-zero full-ResNet18 systems composition using all 21 linear records and the exact 62-item stock stream; and (8) a manifest-bound clean-clone/container gate that rebuilds the paper and canonical checks while rejecting source mutation and private-artifact survivors. A final same-HEAD clean-clone execution remains a release gate.

The corrected three-product Phase C is a local compatibility bug fix, not an independent novelty contribution. The canonical audit records chronology and conceptual overlap with a private project’s earlier correction of a leaky DPF output layer; professor/contributor credit, permission, disclosure, and private-project-overlap rulings remain explicitly open and must be settled before external circulation. The point-DPF and Ring-LPN generator primitives are prior work, not claimed as novel. This is feasibility evidence: the controlled CNN2/CNN3 setup-included preprocessing/dealer ratios of summed layer means are $882.3274\times$ and $60.5768\times$, respectively. Together with unprotected post-handshake local transport and unpinned Ring-LPN parameters, these negative results are publication risks rather than competitive-system evidence.

3 Motivation: why dealerless, and why on GPUs

3.1 Why remove the dealer

The dealer is the single point of *total, retroactive* compromise: it holds every mask of every computation, so one breach — or one subpoena, or one insider — retroactively breaks the privacy of everything the two parties ever computed. The trust assumption is also operationally awkward: in many two-party deployments (two mutually distrusting companies; a hospital and an ML provider) no mutually trusted third machine exists, and dealer key material for training runs is measured in gigabytes that must be generated, stored, and shipped securely. Removing the dealer replaces an institutional assumption with a cryptographic one.

3.2 Why pseudorandom correlation generators, and which one

A training run consumes correlated randomness for billions of secure multiplications; any protocol paying per-multiplication communication in the offline phase is dead on arrival at that scale. Pseudorandom correlation generators (PCGs) [5, 6] restructure the cost: the parties run one small interactive setup yielding short correlated seeds, then each party *locally* — with zero communication, “silently” — expands its seed into millions of correlated samples.

The atomic correlation we need is *oblivious linear evaluation* (OLE): P_0 holds u , P_1 holds v , and they obtain additive shares of $u \cdot v$. Two OLEs make one Beaver triple; Beaver triples are exactly what Orca’s linear-layer keys package. The PCG we build on is the ring-OLE generator of

Boyle et al. [6], which we refer to as the **BCG+20 generator**.¹ In outline, over the negacyclic ring $\mathcal{R}_p = \mathbb{Z}_p[X]/(X^n + 1)$:

1. Each party σ samples c *sparse* secret polynomials $e_0^\sigma, \dots, e_{c-1}^\sigma$ (only t nonzero coefficients each). The public vector is $a = (1, a_1, \dots, a_{c-1})$, with uniform $a_i \in \mathcal{R}_p$ for $i \geq 1$, and its generator input is $x_\sigma = e_0^\sigma + \sum_{i=1}^{c-1} a_i e_i^\sigma$ — pseudorandom under the exact module-Ring-LPN assumption (§4.8).
2. The target product expands as $x_0 x_1 = \sum_{i,j} a_i a_j (e_i^0 e_j^1)$: all secrecy concentrates in the c^2 products of sparse polynomials, and *sparse times sparse stays sparse* — each product has at most t^2 nonzeros, at positions that are *sums* of the two parties’ secret positions with values that are *products* of their secret values.
3. Compact shares of such sparse vectors are exactly what sums of distributed point functions provide (SPFSS, built from DPF trees [3]): kilobyte-sized keys whose full-domain evaluations sum to the hidden sparse vector.
4. Each party locally evaluates its trees everywhere, multiplies by the public $a_i a_j$ (fast NTT polynomial arithmetic), and accumulates. Output: a *ring OLE*, $z_0 + z_1 = x_0 x_1$ in $\mathcal{R}_p$.

Slot packing — the amortization that makes this affordable. Our primes are chosen so that $X^n + 1$ splits into n linear factors mod p ; then the negacyclic number-theoretic transform (NTT) is a ring *isomorphism* $\mathcal{R}_p \rightarrow \mathbb{Z}_p^n$, and reading one ring OLE per coordinate (“slot”) yields up to $n = 8,192$ *independent scalar OLEs* from a single ring-OLE expansion. Because the NTT is linear, each party transforms its own shares locally. In the measured isolated-expansion artifact (§5.3), suite shapes consume $2L \lceil MKN/n \rceil$ ring OLEs (L is the limb count) — e.g. **two** for a $16 \times 32 \times 16$ layer at $Q = p_0$, instead of 16,384 per-scalar OLE calls. A self-sustaining factory must reserve next-epoch keygen correlations, reducing usable capacity to at most $n - 3c^2 t^2$ and increasing this count (§4.3).

3.3 Why GPUs

The motivation is empirical, not aesthetic. Profiling the expansion (§5.3) shows that **SPFSS full-domain evaluation — millions of independent AES/PRG invocations across thousands of DPF trees — dominates the cost** ($\sim 99\%$ at realistic noise weights; the NTT is under 1%). That workload is embarrassingly parallel and exactly GPU-shaped, and it is why our expansion runs in milliseconds on one GPU (9.086 ms q64 and 18.230 ms q128 at the regular $t=8$ smoke parameters, one iteration per configuration; 61–881 ms at $t=64$ depending on noise regime, with two timed iterations per mode). The live distributed key generation (§4.3) preserves this shape: all trees advance *level-synchronously*, so the depth-14 tree walk has 14 sequential batches independent of how many thousands of trees are in flight. This is not an end- to-end network-round count. The implementation reports the input adder, correction-word openings, and two-stage payload multiplications as semantic dependency layers, separately from transport direction switches. Orca itself is GPU-resident; its randomness factory should live on the same hardware, and our keys are written byte-compatibly so Orca’s GPU online path runs unmodified.

¹The construction is specified in Figure 2 of [6]; our codebase and earlier internal reports call the implementation the “Figure-2 engine” after that figure number. The present document uses “BCG+20 generator” to avoid referencing a figure that does not appear in it.

3.4 Notation

M, K, N	linear-layer dimensions ($M \times K$ by $K \times N$ matmul)
bw	Orca’s ring width; shares live in $\mathbb{Z}_{2^{\text{bw}}}$ ($\text{bw} \leq 32$ here)
n	ring dimension in $\mathcal{R}_p = \mathbb{Z}_p[X]/(X^n + 1)$; the source manifest uses 8192, while the retained full-graph control overrides it to 262144; neither is security-pinned
p_0, p_1	current artifact primes $2^{62} - 6 \cdot 2^{24} + 1$, $2^{62} - 7 \cdot 2^{24} + 1$; not pinned
L, Q	CRT limbs; $Q = p_0 p_1$ ($L=1$: “q64”, $Q = p_0$; $L=2$: “q128”)
d_{DPF}	DPF depth ($\log_2(2n)$ for the uniform-noise artifact)
c, t	Ring-LPN noise parameters: c sparse polynomials, weight t each
λ	target symmetric parameter 128; GPU expansion now uses full 128-bit seeds with separate control-bit PRF outputs

4 Live dealerless forward-FC design on GPUs

4.1 Design overview and principles

Figure 2 shows the composed live forward path. Every stage is implemented and gate-verified for the stated feasibility configurations. Components D1–D4 implement the earlier centralized-keygen, conversion, randomness, and process-separation boundaries for this scope; this does not close P-KEY or the reopened P-CONV proof obligation. D5 remains open. Three principles govern the artifact:

1. **The online path is untouchable.** Keys are written byte-compactly; Orca’s online matmul (`gpuMatmulBeaver`) is both the consumer and the independent validation oracle. (With the integration flag off, baseline Orca is byte-identical — verified.)
2. **Oracle boundaries are explicit.** Every idealized component is marked at the exact line in the source and named in every report; numbers produced with an oracle in the loop are never mixed with protocol-backed numbers in one table column.
3. **Every claim has a mechanical gate.** Components ship with suites that exit non-zero on any failure; one command re-validates the whole chain (§5.2).

4.2 Setting and threat model

Two computationally bounded parties; **static semi-honest** (honest-but-curious) corruption of one party. The conditional theorem assumes authenticated point-to-point channels, AES as a PRG/PRF ($\lambda=128$), standard semi-honest security of the selected OT backend, the Gilboa field-multiplication wrapper, and an actual-map BGI DPF reduction including the leaf-conversion statistical loss, plus decisional splittable Ring-LPN for the exact selected sparse-noise distribution (§4.8) and a complete output-conditioned conversion simulator. These proof obligations remain open. The canonical and classifier suites instantiate SCI/IKNP. The separate EMP-Silent correctness/accounting rerun remains opt-in and independently unreviewed. Current measurements use same-host loopback after a mutual HMAC-SHA256 endpoint/context handshake, but later local protocol bytes have no application-layer integrity. They are implementation/correctness evidence for the conditional theorem rather than an authenticated-channel deployment.

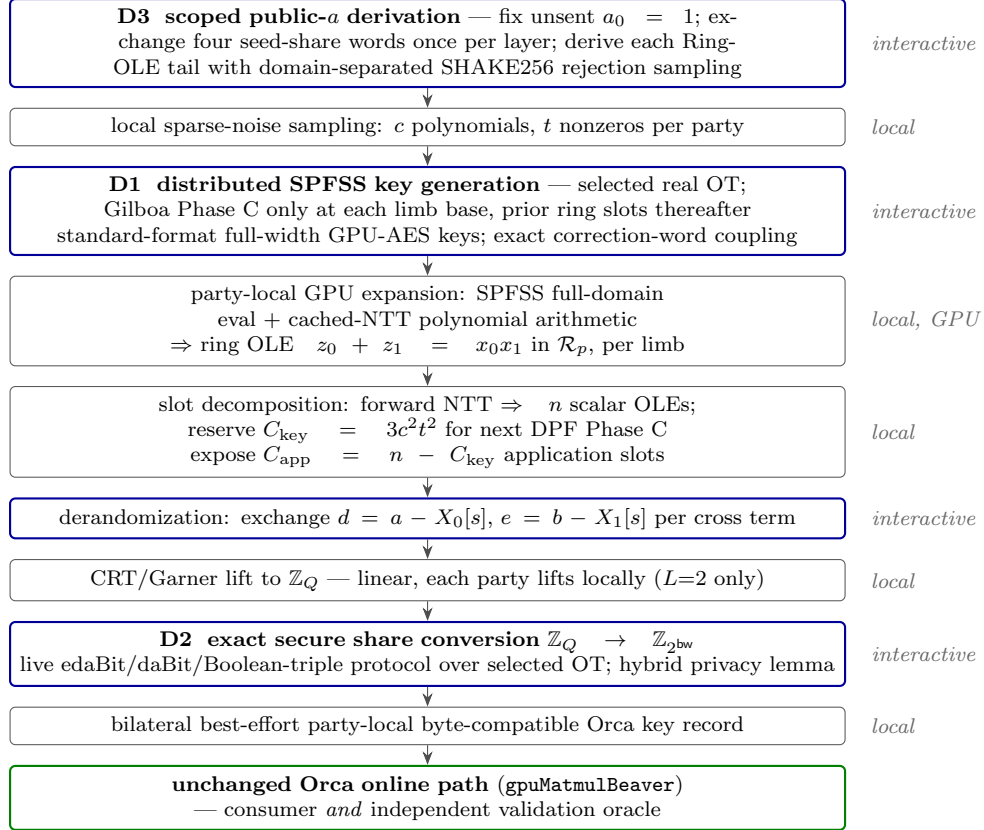


Figure 2: Live forward-linear pipeline. Both parties run as independent processes on distinct GPUs. Blue-framed stages exchange messages over same-host TCP after mutual HMAC-SHA256 endpoint/context establishment; later protocol bytes have no per-message integrity. Other stages are party-local, and Ring-LPN expansion runs on the GPU. The unchanged online consumer runs only in the post-exit checker.

Out of scope: malicious security, external packet attackers, timing and microarchitectural side channels, stateful training transitions, and dealerless nonlinear DCF preprocessing. “Dealerless Orca” without the forward-linear/FC qualifier is not claimable.

Current evidence boundary. The host artifact of §5.3 remains a counted ideal-functionality prototype using the noncryptographic splitmix64 correctness PRG. The live path separately uses full-width GPU AES, the selected real OT backend, epoch-zero Gilboa OLE, consume-once Ring-OLE-output bootstrap, a four-word joint public seed, domain-separated SHAKE256 public-polynomial derivation, and separate process/GPU state. The canonical and classifier suites use SCI/IKNP. The separate current EMP-Silent route supplies focused correctness and accounting evidence only; it remains independently unreviewed. The proof models SHAKE256 as a public random oracle and remains conditional on the unpinned Ring-LPN assumption; measured local traffic is not an authenticated deployment. Therefore neither backend artifact establishes an end-to-end 128-bit-security claim.

4.3 D1: Distributed SPFSS key generation (removes G1)

What must be computed. For each pair (k, l) of noise nonzeros, the parties must generate DPF keys for the point function with position $\alpha = \text{pos}_k^0 + \text{pos}_l^1$ (a value in $[0, 2n)$: products of degree- $< n$ polynomials have unreduced degree up to $2n - 2$; the negacyclic fold $u[i] - u[i + n]$ is linear and applied locally after expansion) and payload $\beta = \text{val}_k^0 \cdot \text{val}_l^1 \in \mathbb{Z}_p$. Observe that the required sharing is *already present in the problem*: α is additively shared (each party knows its own summand) and β is multiplicatively shared, with no extra protocol.

Protocol. We adopt the Doerner–shelat two-party DPF key generation [18] in the variant used by the PCG literature [5, 6]. The parties walk the GGM tree level by level. Off the secret path the two parties’ nodes are identical, so XORing each party’s aggregate PRG expansions leaves the on-path contribution needed for that level’s correction word. Selection by the current secret-shared bit of α uses a 1-of-2 OT on λ -bit strings. We keep control/tag bits separate from the full λ -bit seed state, following the formal BGI construction [4]. We do not derive tags from seed LSBs: the S1 target carries 128 secret seed bits and separate tags.

For the arithmetic payload, let A_b be party b ’s signed leaf-seed aggregate and F_b its signed control-bit aggregate. A first scalar OLE produces $\gamma_0 + \gamma_1 = \beta_0 \beta_1 = \beta$. Set

$$d_0 = \gamma_0 - A_0, \quad d_1 = \gamma_1 - A_1, \quad s_0 = F_0, \quad s_1 = F_1,$$

so $d_0 + d_1 = \beta - A_0 - A_1$ and $s_0 + s_1 = F_0 + F_1 \in \{+1, -1\}$. Two additional directional OLEs share $d_0 s_1$ and $s_0 d_1$; adding those cross-term shares to the local products $d_0 s_0$ and $d_1 s_1$ yields shares $w_0 + w_1 = (d_0 + d_1)(s_0 + s_1)$. The parties open only this standard public key material, $\text{finalCW} = w_0 + w_1$; they never open the d_b , s_b , or their sign. The predecessor transcript opened both F_b values. Although their sign is marginally independent of α , conditioning on one party’s leaf control-bit vector makes it select the class containing α , so that opening was invalid.

The input bits of α come from a secure ripple adder ($\log(2n) - 1$ Beaver bit triples). This is the concrete integration choice for arithmetic position shares, not a claim of new ripple-adder research. The known 1-OT-per-level walk remains an unimplemented performance optimization.

Prototype status (implemented; §5.3). The party-separated host implementation generates standard-format DPF keys from shared (α, β) and passes full-domain validation by the *unchanged* existing evaluator on every tree tested: 2,432 key pairs, both CRT primes, depths through $\log(2n) = 14$. It uses ideal-functionality OT, bit triples, and three scalar OLEs, plus the evaluator’s non-cryptographic correctness PRG; these are exactly why the executable is not evidence of computational privacy. At depth 14 it records 28 string OTs, 13 bit triples, 3 scalar OLEs, 1,908 logical opened bits, and 3,816 meaningful share bits per tree. Neither bit count is measured transport traffic. The old-sign regression reconstructs the removed sign only in the omniscient harness and confirms that it selects a proper leaf-control-bit class containing the true point; this is a regression model of the old leak, not a privacy proof.

A real-transport path runs the same protocol as two OS processes over SCI/IKNP and Gilboa OLE: 369/369 host-reference keys pass the unchanged evaluator; 88 full-width-AES keys pass batched and per-tree GPU evaluation. The exact coupling lemma below proves that its correction words equal conditioned standard DPF generation, so functional equality is no longer substituted for distribution equality. Single-key privacy still invokes the standard DPF/PRG theorem; this component result does not establish Ring-LPN security.

4.4 S1 security contract for D1 and FC composition

Two-level ideal functionality, freshness, and state topology. The numeric *sid* remains a compatibility/session handle. The actual global base invocation is a public nonzero 128-bit *iid*. For every primitive the parties use the fixed-width big-endian tuple

$(v, iid, layerID, kind, layer, direction, limb, ringBatch, tree, phase, ordinal, convChunk, outputSlot),$

with $2^{64} - 1$ in every nonapplicable unsigned coordinate, and hash the domain-prefixed encoding with SHA-256. The encoding is injective before hashing. Hash collision resistance is an explicit assumption. Here *layerID* hashes only the stable operator, shape, arithmetic, Ring-LPN parameters, and layer ordinal; it excludes *iid*, compatibility SID, ledger, deployment channel, and OT backend. Before OT setup or private-DRBG construction, both parties validate the exact finite range plan and consume *iid* in owner-only append-only ledgers: no-replace create, write/fsync, atomic rename, and directory fsync. A duplicate, retry/restart, collision, malformed/truncated or crash-left pending entry aborts before publication; consumed attempts never roll back and unused tail slots are discarded. Both version-3 party records carry *iid* and the digest binding it to the layer and plan. This additionally assumes a trusted local filesystem with the stated durability and no adversarial storage rollback. All publication attempts in one administrative deployment must use this same persistent namespace; replacing it with a fresh, copied, or rolled-back root is an explicit excluded deployment behavior.

For a public $(iid, cid, tree, d_{DPF}, p)$, $\mathcal{F}_{DDPF}$ receives $(off_b, \beta_b, root_b)$ from P_b , where $0 \leq off_b < 2^{d_{DPF}-1}$, $\beta_b \in \mathbb{Z}_p^*$ is canonically encoded, and $root_b$ is that party's uniform 128-bit random-tape draw. It rejects a malformed, unclaimed, or duplicate identifier with the same public abort at both parties before consuming correlation. Otherwise it sets $\alpha = off_0 + off_1$ without modular wrap and $\beta = \beta_0 \beta_1 \bmod p$, runs standard DPF generation conditioned on the supplied roots, and returns only K_b to P_b . A batch may contain arbitrarily correlated or repeated private inputs, but every tree uses a distinct identifier, independent roots, and fresh primitive correlation.

The stateful $\mathcal{F}_{FC}$ takes a public ordered topology table whose entries reference public mask-wire handles. A handle names one $\mathbb{Z}_{2^{bw}}$ mask and its additive shares; repeated handles intentionally reuse those shares. A training-layer row names $(X, W, Y_{bias}, V_W, V_Y, G)$, its public truncation/optimizer parameters, and whether momentum is enabled; current **FC**Layer requires bias. Each matrix product creates a fresh product-output handle R_C (source-level **mask_Z**) and splits

$$C_0 + C_1 = AB + R_C \pmod{2^{bw}}.$$

Forward reads (X, W) and emits $X_b \| W_b \| C_{f,b}$; weight gradient reads the same X plus the incoming gradient handle G and emits $G_b \| C_{dW,b}$; input gradient, when enabled, reads that same G and the same W and emits only $C_{dX,b}$.

The source-level forward transition broadcasts and adds persistent Y_{bias} to $R_{C,f}$ before truncation; the truncated result is the next activation-mask handle. Backward row-sums G to obtain the bias gradient dY , truncates the dX output, and applies both optimizer transitions $(W, V_W, dW) \mapsto (W', V'_W)$ and $(Y_{bias}, V_Y, dY) \mapsto (Y'_{bias}, V'_Y)$. Current Orca initializes W, Y_{bias}, V_W, V_Y to zero; X comes from the preceding truncated output. These persistent operand/state masks are neither resampled nor assumed independent per invocation. Every OT/DPF/OLE/conversion correlation remains fresh even when a public mask handle is reused. The live artifact implements and checks one complete forward FC matmul. Bias/truncation, backward dW/dX , bias-gradient, and optimizer state transitions require a new stateful extension before any training-layer claim.

All ideal calls carry the complete tuple above. DPF Phase A triples, Phase B directional string OTs, Phase C scalar OLEs, public-polynomial coefficients, derandomization openings, conversion

daBits/edaBits/Boolean triples, and used output slots occupy disjoint coordinates. The transport backend is not a coordinate, so switching IKNP/silent implementations cannot authorize reuse. Repeated operand-mask values remain legal only under distinct correlation IDs. $\mathcal{F}_{\text{BT}}$ returns private XOR shares of fresh uniform $(a, b, c = a \wedge b)$; the Beaver openings remain real protocol messages. $\mathcal{F}_{\text{OT}}^{128}$ reveals only the selected sender string, and $\mathcal{F}_{\text{OLE}}^p$ samples g_0 uniformly and sets $g_1 = x_0x_1 - g_0$. The composition additionally names these boundaries. $\mathcal{F}_{\text{BMUL}}^p$ instead consumes a registered Ring-OLE slot with $Z_0 + Z_1 = X_0X_1$, reveals $\Delta_b = u_b - X_b$, and returns $g_0 = Z_0 + \Delta_1X_0 + \Delta_0\Delta_1$ and $g_1 = Z_1 + \Delta_0X_1$. These output shares retain their exact correlation with the registered slot; they are not fresh-share $\mathcal{F}_{\text{OLE}}^p$ outputs. $\mathcal{F}_{\text{PUBA}}^{p,H}$ exchanges one uniform 256-bit seed share from each party per layer and XORs them into one joint seed. For each fresh Ring-OLE scope it fixes the unsent identity $a_0 = 1$ and rejection-samples the tail from the exact V2 byte query Q defined in §4.6. Distinct full scope/parameter queries define independent exact-uniform $a_1, \dots, a_{c-1}$ in the random-oracle model. $\mathcal{F}_{\text{RINGOLE}}^{x,b}$ is role-indexed and conditioned on the corrupt party’s complete local expansion state. It reruns the source-local expansion to preserve both $X_b = e_{b,0} + \sum_{i=1}^{c-1} a_i e_{b,i}$ and the recomputable Z_b , samples only hidden X_{1-b} uniformly, and sets $Z_{1-b} = X_bX_{1-b} - Z_b$. Thus neither corrupt output is resampled. $\mathcal{F}_{\text{CONV}}$ handles canonical CRT shares. For canonical $z_b \in [0, Q)$, this functionality computes

$$S = z_0 + z_1, \quad \text{wrap} = [S \geq Q], \quad v = S - \text{wrap } Q,$$

then returns uniform additive shares of $v \bmod 2^{\text{bw}}$ and never opens *wrap*. The live composition realizes these interfaces over the selected real OT backend and validates its finalized keys only after both party processes exit.

For canonical additive shares, $a_0 + a_1 < 2^{\text{bw}+1}$ and $b_0 + b_1 < 2^{\text{bw}+1}$, so a length- K integer dot product is strictly below $K 2^{2\text{bw}+2}$. The FC functionality accepts only public triples satisfying $K 2^{2\text{bw}+2} < Q$: reduction modulo 2^{bw} then gives the intended mask product before any reduction modulo Q . This rules out q62/bw32 entirely, caps q62/bw24 at $K \leq 2^{12} - 1$, and is checked before masks are read; q62/bw32 inadmissibility is independently exercised by the executable negative control in `test_orca_zp_bridge`; the live FC path enforces the same public bound before private-mask sampling and never performs a clear value-dependent test. The deployed Ring-LPN GPU expansion uses four domain-separated AES calls per node: plaintexts 0 and 2 produce full 128-bit child seeds, while plaintexts 1 and 3 produce separate control bits. Host/device parity and 88 two-process GPU-evaluated keys are gated. The centralized benchmark kernel still derives roots from one 64-bit `seed_base` and is not security evidence. The live path instead uses OpenSSL-private-DRBG roots/noise/masks and fixes the unsent identity $a_0 = 1$. It exchanges four seed-share words once per layer and derives every per-instance $(c - 1)n$ tail through domain-separated SHAKE256 rejection sampling. This is exact in the stated random-oracle hybrid, not a standard-model public-coin construction. Exact key-distribution coupling does not discharge the actual leaf-map privacy reduction (P-KEY). The corrected conversion inverse is only a necessary local consistency step, not a complete conditional transcript proof (P-CONV). Concrete Ring-LPN security parameters and an authenticated deployment also remain open; no end-to-end 128-bit claim is made.

Leakage and abort contract. Common leakage consists of protocol/session and party-role identifiers; the parameter set, $L, d_{\text{DPF}}, p, n, c, t$, CRT limbs, noise mode, epoch and batch/tree counts; public layer shapes, layouts, invocation kind, wire-handle topology, slot/access order, serialized field lengths, fixed message lengths and dependency schedule; common DPF correction words and *finalCW*; masked openings ϵ, δ, d, e and the specified conversion openings; both exchanged 256-bit public-seed shares, their XOR joint seed, and the public Ring-LPN vectors derived from it; and

the common accept/abort stage. P_b additionally sees only its own DPF inputs/root draw, retained mask shares, local frontier/noise/output-mask state, sender/receiver OT view, triple/OLE shares, authenticated messages, output DPF keys, and emitted Orca fields. The other inputs, roots, keys, mask shares, hidden leaf-control sign, conversion wrap, unused OT messages, OLE masks, and PCG state remain hidden. Public mask-handle reuse is required topology; reuse of primitive correlation or private random-tape draws is prohibited. External packet observers, timing/microarchitectural side channels, active security, fairness, and availability remain out of scope.

Exact D1 hybrid transcript. The proof target is the $(\mathcal{F}_{\text{BT}}, \mathcal{F}_{\text{OT}}^{128}, \mathcal{F}_{\text{PHASEC}}^p)$ -hybrid protocol instantiated by the current source, where each Phase-C product uses fresh-share $\mathcal{F}_{\text{OLE}}^p$ at a limb's base epoch or consume-once stateful $\mathcal{F}_{\text{BMUL}}^p$ thereafter. *Phase A*: for each of the $d_{\text{DPF}} - 1$ carry positions, obtain one fresh bit triple, reveal both parties' shares of the two masked one-bit values (ϵ_j, δ_j) , apply the Beaver equation, and update XOR-shared carry and point bits. This exposes $2(d_{\text{DPF}} - 1)$ logical bits as $4(d_{\text{DPF}} - 1)$ transmitted shares. The private position summands are the Ring-LPN noise positions, so uniform-mode sums are triangular; regular-mode offsets give the corresponding public-bucket shift, not a uniform modular point.

Phase B: at level i , each party locally expands its live seeds, XOR-aggregates left/right seed and flag children, and defines $Z_b = S_b^L \oplus S_b^R$. Two directional 128-bit OTs share $a_i(Z_0 \oplus Z_1)$. Parties reveal their shares of the standard seed correction word and two flag correction words, then advance locally. These are exactly $130d_{\text{DPF}}$ logical opened bits and $260d_{\text{DPF}}$ meaningful share bits.

Phase C: one OLE shares $\gamma_0 + \gamma_1 = \beta_0\beta_1$. With $d_b = \gamma_b - A_b$ and $s_b = F_b$, two directional OLEs share d_0s_1 and s_0d_1 . Each party sets $w_b = d_bs_b + x_b + y_b$ and reveals only w_b , giving the standard $\text{finalCW} = w_0 + w_1$. Neither d_b , s_b , nor $s_0 + s_1$ is opened. Thus

$$\begin{aligned} \text{logical opened bits} &= 2(d_{\text{DPF}} - 1) + 130d_{\text{DPF}} + \lceil \log_2 p \rceil, \\ \text{meaningful share bits} &= 4(d_{\text{DPF}} - 1) + 260d_{\text{DPF}} + 2\lceil \log_2 p \rceil. \end{aligned}$$

Concrete post-base realization: a reserved prior Ring-OLE slot gives (X_b, Z_b) with $Z_0 + Z_1 = X_0X_1$. For any Phase-C product u_0u_1 , the parties exchange $\Delta_b = u_b - X_b$ and output

$$g_0 = Z_0 + \Delta_1X_0 + \Delta_0\Delta_1, \quad g_1 = Z_1 + \Delta_0X_1.$$

Their sum is u_0u_1 . In the role-conditioned ideal Ring-OLE boundary the honest X_{1-b} is uniform, hence the received Δ_{1-b} is a one-time pad independent of the honest input. The simulator samples that difference uniformly and computes the corrupt output with the same role-specific equation and retained (X_b, Z_b) state. This exactly realizes $\mathcal{F}_{\text{BMUL}}^p$; it does not claim a fresh-uniform OLE output share. Distinct consume-once slot identifiers compose sequentially. The abstract counts above remain the standalone DPF contract. A bootstrapped tree concretely opens six masked field elements plus finalCW , recorded as $7\lceil \log_2 p \rceil$ logical and $8\lceil \log_2 p \rceil$ transmitted-share-width bits in Phase C.

Exact coupling to standard DPF generation. After any fixed common-CW prefix, pair the parties' states at each tree prefix. The standard invariant gives equal seed/tag states off the path of α and opposite tags at its prefix, so all off-path values cancel from the joint XOR aggregates. With $a = a_0 \oplus a_1$ and $Z_b = S_b^L \oplus S_b^R$, substituting the two selected OT outputs into the opened seed-CW shares gives

$$sCW = S_0^R \oplus S_1^R \oplus a(Z_0 \oplus Z_1) = \begin{cases} S_0^R \oplus S_1^R, & a = 0, \\ S_0^L \oplus S_1^L, & a = 1. \end{cases}$$

This is exactly the standard losing-branch seed correction. The flag words are

$$tLCW = T_0^L \oplus T_1^L \oplus a \oplus 1, \quad tRCW = T_0^R \oplus T_1^R \oplus a,$$

so induction preserves the invariant and matches every standard correction word. At the leaves cancellation gives $A = A_0 + A_1 = \text{convert}(s_{0,\alpha}) - \text{convert}(s_{1,\alpha})$ and $F = F_0 + F_1 = t_{0,\alpha} - t_{1,\alpha} \in \{+1, -1\}$. Phase C reconstructs

$$\text{finalCW} = (\beta - A)F,$$

which is $\beta - A$ for tags $(1, 0)$ and $A - \beta$ for $(0, 1)$, exactly the standard final correction. Thus, conditioned on the supplied roots, both output keys equal standard generation with the same leaf map deterministically; independent uniform roots give the same joint distribution. This algebraic coupling does not prove single-key privacy for the actual PRG and leaf map.

P-01: ideal-leaf bias and the open P-KEY reduction. Both host and GPU `convert_zp` sum the separately reduced 64-bit halves of a 128-bit seed modulo p . Even with ideal independent uniform halves this is not exact-uniform in $\mathbb{Z}_p$. Write $2^{64} = kp + r$, $0 \leq r < p$. Each half has $k + [v < r]$ preimages; their modular sum has $k^2p + 2kr + T(v)$, where T is the cyclic convolution of the interval indicator on $[0, r - 1]$ with itself and $\sum_v T(v) = r^2$. Thus the distribution is

$$(1 - \epsilon)U_p + \epsilon D_r, \quad \epsilon = \frac{r^2}{2^{128}}, \quad \Delta(\text{Convert}(U_{128}), U_p) \leq \epsilon,$$

where D_r is the modular sum of two uniform values in $[0, r - 1]$ (for $r = 0$ the map is uniform). This is a bound per ideal-uniform leaf, not a complete DPF-security bound.

For $p = 4611686018326724609$, $k = 4$ and $r = 402653180$. Here T is triangular on $[0, 2r - 2]$, and

$$\epsilon = \frac{10133098960257025}{21267647932558653966460912964485513216}, \quad \log_2 \epsilon \simeq -70.830075.$$

A shift by $\lfloor p/2 \rfloor$ makes the excess supports disjoint. For candidate payloads at the same known α , a party-0 key computes its own leaf tag. In the ideal-leaf hybrid, when that tag is 1, evaluation at α equals $\beta + \text{Convert}(s_{1,\alpha})$. The shifted-interval event has probability gap ϵ conditional on that tag, or $\epsilon/2$ including the independent uniform tag event. The 8-bit-half analogue at $p = 61$, shift 30, and interval $[0, 22]$ has exact count gap 144 among 256^2 pairs. This same- α payload-shift witness is *not an empirical AES attack*, nor the full P-KEY reduction.

Qualified review must justify the actual AES/DPF ideal-leaf replacements and their number, then compose these statistical terms with the computational losses over keys, both CRT limbs, directions, batches, epochs, and layers for a declared lifetime. A sum of the applicable per-leaf bounds is only one component after those replacements are justified, not the final concrete advantage budget. This correction changes no primitive and makes no 128-bit claim.

D1 batch-simulation lemma at the ideal boundary. Fix either corrupt party, its complete arbitrarily correlated input/root vector, public order and identifiers, and its full ideal output-key vector. Condition each key on the same root supplied by that party. The following algorithms generate the complete hybrid view with the real conditional distribution. The exact coupling above identifies those conditioned outputs with standard DPF generation; concrete PRG/transport security is separate.

Corrupt P_0 . For each tree, take the root seed and common correction words from K_0 and expand the local frontier. At every carry, sample P_0 's three bit-triple shares, compute its actual

sent shares of (δ_j, ϵ_j) , sample the two reconstructed openings uniformly, and set the honest sent shares to the XOR complements. Apply P_0 's real Beaver equation, including the public $\delta_j \epsilon_j$ term, and the carry recurrence. Hidden honest triple shares one-time-pad both openings, so induction from carry zero gives the exact joint carry/point-share view.

At every level, derive P_0 's aggregates. In the first OT it is receiver: sample the selected output, uniform under the honest sender's fresh mask. In the second it is sender: sample r_0 and record $(r_0, r_0 \oplus Z_0)$. Compute its seed/flag opening shares with the real equations and set the honest shares to complements of the words fixed by K_0 . Use the following source branch for each of the three Phase-C products. For a declared product with output ρ and inputs (u_0^ρ, u_1^ρ) , at a limb's base epoch sample the corrupt $\mathcal{F}_{\text{OLE}}^p$ output ρ_0 uniformly. At a later epoch, take that product's retained slot (X_0^ρ, Z_0^ρ) , set $\Delta_0^\rho = u_0^\rho - X_0^\rho$, sample only the honest masked difference Δ_1^ρ uniformly, and compute

$$\rho_0 = Z_0^\rho + \Delta_1^\rho X_0^\rho + \Delta_0^\rho \Delta_1^\rho.$$

Apply this branch first to $(u_0^\gamma, u_1^\gamma) = (\beta_0, \beta_1)$, then set $d_0 = \gamma_0 - A_0$ and $s_0 = F_0$. Apply it next to $(u_0^x, u_1^x) = (d_0, s_1)$ and $(u_0^y, u_1^y) = (s_0, d_1)$. Thus (γ_0, x_0, y_0) need not be independent or uniform after the base epoch. Compute $w_0 = d_0 s_0 + x_0 + y_0$ and emit only $w_1 = \text{finalCW} - w_0$. This ordering also covers zero intermediate values and preserves the complete retained-state correlation; it does not infer the hidden sign.

Corrupt P_1 . Take the root and common words from K_1 . Sample its triple shares and masked openings as above, but apply P_1 's Beaver equation without the public $\delta_j \epsilon_j$ term. At each level record its first-OT sender pair $(r_1, r_1 \oplus Z_1)$ and sample its selected second-OT receiver output; derive its own seed/flag shares and complement to the fixed common words. Use the same source branch for each declared product. At a limb's base epoch, sample the corrupt $\mathcal{F}_{\text{OLE}}^p$ output ρ_1 uniformly. At a later epoch, take that product's retained (X_1^ρ, Z_1^ρ) , set $\Delta_1^\rho = u_1^\rho - X_1^\rho$, sample only the honest masked difference Δ_0^ρ uniformly, and compute

$$\rho_1 = Z_1^\rho + \Delta_0^\rho X_1^\rho.$$

Apply this branch first to $(u_0^\gamma, u_1^\gamma) = (\beta_0, \beta_1)$, then set $d_1 = \gamma_1 - A_1$ and $s_1 = F_1$. Apply it next to $(u_0^x, u_1^x) = (d_0, s_1)$ and $(u_0^y, u_1^y) = (s_0, d_1)$. Thus (γ_1, x_1, y_1) need not be independent or uniform after the base epoch. Compute $w_1 = d_1 s_1 + x_1 + y_1$ and emit only $w_0 = \text{finalCW} - w_1$, preserving the complete retained-state correlation.

Both simulators enforce a consumed-correlation-ID ledger before every ideal call and proceed tree by tree conditioned on the complete prior state. Therefore repeated/correlated private summands do not invalidate the one-time-pad hybrids. S1 treats local coins as independent consumed random-tape draws and makes the root explicit; S3 must prove a concrete, state-consistent CSPRNG interface. In the end-to-end hybrid, only the honest party's hidden PRG stream may be replaced by ideal random strings; the corrupt party's stream and retained state remain consistent.

Conditional forward-FC proof target. Let χ_U sample exactly t distinct positions without replacement and χ_R sample one position in each of t public equal buckets; in both cases coefficients are independent uniform elements of $\mathbb{Z}_p^*$. Assume decisional module-Ring-LPN for the exact selected distribution: for independent uniform $a_1, \dots, a_{c-1} \in R = \mathbb{Z}_p[X]/(X^n + 1)$ and $e_0, \dots, e_{c-1} \leftarrow \chi$,

$$(a_1, \dots, a_{c-1}, e_0 + \sum_{i=1}^{c-1} a_i e_i)$$

is indistinguishable from replacing the final component by uniform R . Sampling all c public entries would be a different, unreviewed assumption. Also model SHAKE256 on the exact V2 byte query

ID	Proof obligation	Current status
P-CORR	Keys reconstruct $\beta[x = \alpha]$.	2,432/2,432
P-DIST	Couple correction words to standard DPF generation conditioned on both roots.	executable passes
P-KEY	Hide point/payload under the actual AES expansion and biased leaf map.	exact algebraic coupling above
P-POS	Prove the exact uniform/regular Ring-LPN exponent distribution.	open: P-01 statistical loss and lifetime reduction/composition contract fixed; distribution/reduction audit open
P-ADD	Simulate either ripple-adder view with explicit triple/opening equations.	closed in bit-triple hybrid
P-LEVEL	Simulate both OT roles and CW shares conditioned on each full key.	closed in OT hybrid
P-PAYLOAD	Simulate the three-OLE view without the removed sign, including zero intermediates.	closed in OLE hybrid
P-BATCH/P-FRESH	Preserve correlated tree inputs and reject correlation-ID reuse before output.	D1 lemma + persistent invocation/ledger controls
P-RNG/P-PCG	Realize private tapes/exact public a and prove Ring-LPN OLE at the selected distribution.	RNG implemented; PCG reduction/parameters open
P-CONV	Realize exact modulo- Q conversion without revealing wrap.	reopened: local inverse corrected; full conditional transcript proof open
P-TOPO/P-PROC	Match stateful reuse/serialization and two-process execution.	forward FC and classifier evidence closed on SCI/IKNP; training state and authenticated deployment open

Table 1: S1 proof contract. Closed rows are hybrid-transcript lemmas only, not a computational-security claim; open rows are binding later-stage gates.

Q of §4.6 as the public random oracle used for independently scoped public- a vectors, and assume the four-call GPU AES expansion is a PRG, a reviewed single-key reduction for the actual leaf map including P-01 statistical loss and lifetime composition, semi-honest security of the selected OT backend and Gilboa field-multiplication wrapper, authenticated channels, the role-indexed BCG+20 Figure-2 reduction, and a complete state-consistent output-conditioned P-CONV simulator. The actual-map reduction, Figure-2 bridge, and full conversion simulator are not discharged here.

Subject to these open obligations, the proposed conclusion is realization of the forward-matmul restriction of $\mathcal{F}_{FC}$ with the leakage above and the composed advantage losses. The conditional proof outline samples one joint 256-bit public seed, sets the honest four-word XOR share against the corrupt share, and programs every fresh exact Q query with an independent uniform stream. It retains the corrupt random tape, local e_b , and D1 key, reruns the source-local expansion, and thereby preserves both $X_b = e_{b,0} + \sum_{i=1}^{c-1} a_i e_{b,i}$ and recomputable Z_b . It then replaces the first limb instance’s external transports by ideal OLE and applies the complete D1 simulator and standard-key coupling; replaces only hidden X_{1-b} and sets $Z_{1-b} = X_b X_{1-b} - Z_b$; then registers that ideal output’s reserved suffix with $\mathcal{F}_{BMUL}^p$. For the next DPF, it samples each honest masked difference uniformly and derives the corrupt Phase-C output from the retained slot exactly. The generalized D1 simulator and Figure-2 replacement then provide the next ideal Ring-OLE state. Induction in public epoch order covers every later DPF and expansion without assuming its own output. It samples each opposing application derandomization opening under the corresponding fresh hidden

OLE share, applies the complete conversion simulator still required by P-CONV, and adds the fresh uniform output-mask share before serializing $A_b||B_b||C_b$. Every replacement must preserve prior state. Across m fresh direction/limb/batch instances, the intended hybrid sums the m role-indexed Figure-2/module-Ring-LPN advantages; the full bound must additionally charge the actual-map DPF statistical/computational losses and justify conversion composition. A joint claim cannot follow from one unconditioned sample. This theorem excludes the measured post-handshake TCP transport, which has no per-message integrity, as well as malicious parties and unimplemented training state transitions. The source execution uses $(n, c, t) = (8192, 2, 8)$; the separate full-graph control overrides $n = 262144$ at the same (c, t) . Both are feasibility configurations with no concrete-security claim.

Bootstrapping (and why it is not circular). Key generation consumes three scalar OLEs per DPF tree; expansion produces n slot OLEs per ring OLE. The factory therefore feeds itself precisely when

$$3c^2t^2 < n, \quad (1)$$

The live protocol realizes these epochs. It exposes the prefix $C_{\text{app}} = n - C_{\text{key}}$ of every Ring-OLE output to FC and reserves the final suffix $C_{\text{key}} = 3c^2t^2$ for the next DPF; it rejects $C_{\text{key}} \geq n$ before OT setup. The first instance of each CRT limb uses a fixed external batch of Gilboa multiplications. In public (*ring batch, direction, limb*) order, every later instance consumes the preceding output's reserved suffix using the stateful masked-difference realization above. The final per-limb continuation reserve is discarded with the consume-once invocation. At $(c, t, n) = (2, 8, 8192)$, $C_{\text{key}} = 768$, $C_{\text{app}} = 7424$, and the output/input surplus is $8192/768 = 10.67\times$.

Noncircularity is an induction: epoch zero first replaces external OLE by $\mathcal{F}_{\text{OLE}}$, then DPF by $\mathcal{F}_{\text{DDPF}}$ and expansion by the role-indexed ideal Ring-OLE. Its reserved suffix registers state for $\mathcal{F}_{\text{BMUL}}$, whose exact role-specific simulator supplies the next DPF's Phase-C view without replacing the correlated output by fresh $\mathcal{F}_{\text{OLE}}$ shares. Repeat the DPF and Figure-2 replacements in public order. Distinct slot IDs are necessary for sequential composition. The conclusion remains conditional on the role-indexed multi-instance Ring-LPN and actual-map P-KEY obligations; epoch order alone does not prove the required retained-state reduction or discharge P-01.

For the classifier run, the exact count is $2L[MKN/C_{\text{app}}] = 276$ Ring-OLE instances ($L = 2$, 69 batches). Per party, 1,536 Phase-C products come from epoch zero, 210,432 come from PCG output, and the 211,968 reserved slots split exactly into 210,432 consumed plus 1,536 terminal-discarded slots; 1,024 application slots are also unused and reported. If $t = 64$ at $n = 8192$, then $3c^2t^2 = 49,152 > 8,192$ and the executable capacity control rejects before output; a reviewed parameter set must increase n .

GPU batching. All trees of a generation batch advance level-synchronously: the depth-14 tree walk has 14 sequential batched AES/PRG expansions and OT selections, independent of tree count. One CUDA block now owns each tree's XOR and final modular reductions; warp/block reductions replace the former same-address leaf atomics without changing the DPF transcript. The secure ripple adder, correction-word openings, and two-stage payload multiplication add dependency stages. The TCP implementation reports semantic dependency layers separately from direction switches; neither counter is a packet/network-round measurement.

P-03: exact live primitive inventory (one ring-OLE limb instance). Let $B = c^2t^2$ and let the tree depth be $d = \log_2(2n)$ for uniform noise or $d = \log_2(2n/t)$ for regular noise. Regular product points are grouped on $c^2(2t - 1)$ public bucket diagonals, but key generation still uses B

DPF trees. At every level the live `ot_duplex_128` sends and receives on two independent directional contexts. The exact Phase-B inventory is Bd chosen-message 128-bit string OTs *per direction*, hence $2Bd$ total, not Bd . The one-OT-per-level walk is unimplemented. Phase A additionally consumes $B(d - 1)$ Boolean triples and Phase C consumes $3B$ field products. These are distinct primitive functionalities.

noise, t ($c=2$)	trees B	depth d	string OTs $2Bd$	bit triples	field products
uniform, $t=8$	256	14	7,168	3,328	768
regular, $t=8$	256	11	5,632	2,560	768
uniform, $t=64$	16,384	14	458,752	212,992	49,152
regular, $t=64$	16,384	8	262,144	114,688	49,152

Table 2: Source-derived DPF primitive inventory for one ring-OLE limb instance at $n = 8192$. String OTs count both directions, each with two 128-bit messages and one choice bit; they are not bit-COT counts or measured bytes. At regular $t = 8$, each direction uses 2,816 OTs, totaling 5,632. The $t = 64$ rows are standalone inventory diagnostics, not feasible self-bootstrapping points: $3B \geq n$.

The earlier λd bit-correlation pricing and 1–8-bit silent-OT traffic range are retired, not corrected into a performance estimate. No reviewed reduction equating a chosen-message 128-bit string OT with bit-COT is asserted. This inventory is not a complete transport budget: Phase-A triple generation, setup, chosen-message adjustment/ciphertexts, correction-word openings, framing, and conversion must be accounted separately. Only the first instance of each CRT limb obtains its $3B$ Phase-C products from external Gilboa OLE; later instances consume reserved Ring-OLE slots and exchange the recorded masked field words. Historical measured SCI/IKNP and EMP-Silent byte counters remain separate, immutable backend evidence. Neither those counters nor this corrected inventory provide a matched direct-OT baseline or a new performance estimate.

4.5 D2: PCG-sourced share conversion (removes G2)

The problem. The pipeline holds additive shares over $\mathbb{Z}_Q$ (prime world, where NTTs exist); Orca needs shares over $\mathbb{Z}_{2^{bw}}$ (hardware world, where they do not). Reducing both shares mod 2^{bw} is wrong whenever the integer share-sum wrapped past Q — and that *wrap bit* depends on both shares jointly, so computing it requires an actual secure protocol.

The protocol (validated two-process artifact). For canonical $z_b \in [0, Q)$, let $S = z_0 + z_1 < 2Q$ and $\ell = \lceil \log_2 2Q \rceil$ ($\ell=63$ at q64, 125 at q128). The `edaBit` functionality samples uniform $R \in \mathbb{Z}_{2^\ell}$, an independent uniform arithmetic first share, and independent uniform XOR masks for its bits; the `daBit` functionality analogously shares one random bit over $\mathbb{Z}_2$ and $\mathbb{Z}_{2^{bw}}$. The live artifact realizes one exact `daBit` with one 128-bit OT, composes one exact `edaBit` from ℓ `daBits` (the coefficient-one arithmetic share makes the sum uniform), and realizes each Boolean triple with two one-bit OTs. Under fresh correlations: (1) mask and open $A = (z_0 + z_1 + R) \bmod 2^\ell$; (2) a ripple adder recovers Boolean shares of $S = A - R$; (3) a second adder obtains $w = [S \geq Q]$ from the carry of $S + (2^\ell - Q)$; (4) write a_b for the raw arithmetic share of a fresh `daBit` d , open $h = w \oplus d$, and compute the corrected arithmetic wrap share

$$w_b = \begin{cases} a_b, & h = 0, \\ (1 - b) - a_b, & h = 1 \end{cases} \pmod{2^{bw}};$$

and (5) return $r_b = (z_b - Qw_b) \bmod 2^{\text{bw}}$. Since $w_0 + w_1 = w \bmod 2^{\text{bw}}$, the output sum is $((z_0 + z_1) \bmod Q) \bmod 2^{\text{bw}}$ and w is never opened directly.

Ideal-OT wrapper lemma. These custom correlations are exact constructions in the $\mathcal{F}_{\text{OT}}$ hybrid. For one daBit, P_0 samples bit x and arithmetic mask r , offers $(x - r, 1 - x - r)$ in $\mathbb{Z}_{2^\ell}$, and P_1 selects with bit y . The arithmetic shares sum to $x \oplus y$, matching the local Boolean shares; the unused message and choice remain hidden and either arithmetic share is uniform. Weighting ℓ fresh daBits by 2^j gives the edaBit arithmetic share $R_b = \sum_{j=0}^{\ell-1} 2^j d_{j,b} \bmod 2^\ell$ and its matching XOR-shared bits; freshness makes reconstructed R uniform. For a Boolean triple, each party samples local a_b, b_b and a fresh sender mask, sends $(m_b, m_b \oplus b_b)$ once, receives once with choice a_b , and XORs its local product, sender mask, and receiver output. Expanding the two cross terms gives $c_0 \oplus c_1 = (a_0 \oplus a_1) \wedge (b_0 \oplus b_1)$. For either corruption the simulator retains the corrupt random tape and selected output and chooses only the hidden mask/unused message; fresh identifiers give the batched weighted-edaBit and triple composition used below.

Independent processes over TCP pass 76 conversions in each of four q64/q128 and bit-width configurations: exact boundary sums $0, Q - 1, Q, 2Q - 2$, random inputs, forced wraps, and layer-shaped inner products, with zero mismatches and a corrupted-output control. The ripple circuit consumes $2(\ell - 1)$ AND triples, $2\ell - 1$ post-mask dependency stages, $5\ell - 3$ logical opened bits, and $10\ell - 6$ meaningful share bits. The last count excludes byte padding, framing, setup, and OT payloads; the artifact separately measures setup, correlation, and online bytes and direction switches. Dependency stages are semantic depth, not measured network rounds.

P-02: corrected local inverse; conditional transcript proof open. A is uniform because R is uniform; fresh Boolean triples mask the ripple-AND openings. The B2A opening is $h = w \oplus d$, not w . For either fixed h , the transformation from raw a_b to corrected w_b is a permutation. The fresh uniform arithmetic mask and odd Q therefore give a uniform local output share subject to the required sum.

An output-conditioned local sampler must sample h *first*. For prescribed r_b , compute in $\mathbb{Z}_{2^{\text{bw}}}$

$$w_b = Q^{-1}(z_b - r_b), \quad a_b = \begin{cases} w_b, & h = 0, \\ (1 - b) - w_b, & h = 1. \end{cases}$$

The deterministic source computation then returns r_b in either branch. The earlier inverse incorrectly fixed raw a_b before independently sampling h . At q64/bw3, $Q \bmod 8 = 1$, $b = 0$, $z_b = r_b = 0$, raw $a_b = 0$, and $h = 1$, it returns 7 rather than 0. The corrected local inverse passes exact arithmetic checks for all 256 combinations of role, h , input residue, and prescribed output in $\mathbb{Z}_8$.

Local inversion is not a complete output-conditioned simulator. Qualified review must construct the joint Boolean daBit, edaBit/triple, earlier-opening, local-random-tape, and OT-wrapper state conditioned on r_b and sampled h , preserving every recomputable dependency and the real conditional law. Independently sampling the remaining marginals and completing honest shares does not prove this. P-CONV is therefore reopened even in the $(\mathcal{F}_{\text{edaBit}}, \mathcal{F}_{\text{daBit}}, \mathcal{F}_{\text{BT}})$ hybrid; fresh IDs alone do not close its sequential composition. The inspected source already implements the correct branch. Concrete transport additionally requires the selected backend's semi-honest OT security and authenticated channels. Automated correction is not qualified human security review or circulation authorization.

What remains for production D2. The complete P-CONV proof above is required before a privacy claim. Separate prospective engineering work includes *sourcing*: AND triples come

from a binary PCG (the $\mathbb{Z}_2$ instantiation of our own pipeline, or expand-accumulate codes [20]; Ferret [19] yields them at ~ 1 bit amortized), edaBits from random bit-shares by the standard construction [21] ($O(\ell)$ AND triples each, one-time batch), daBits from the same batch. *Depth*: the ripple comparator becomes a parallel-prefix (Rabbit-style [22]) comparison — carries obey an associative generate/propagate rule, so all of them come out of a $O(\log \ell)$ -depth tree. That depth bounds the eventual interactive schedule, but actual network rounds must be measured after the prefix circuit is implemented and batched.

per $\mathbb{Z}_{2^{bw}}$ output	q64 ($\ell=63$)	q128 ($\ell=125$)
AND triples = $2(\ell-1)$	124	248
edaBit bits = ℓ	63	125
daBits	1	1
logical opened bits = $5\ell - 3$	312	622
meaningful share bits = $10\ell - 6$	624	1,244
post-mask dependency stages, ripple = $2\ell-1$	125	249
post-mask dependency stages, prefix target	~ 7	~ 8

Table 3: Validated two-process conversion cost per output with executable transcript accounting, plus the prefix-comparator dependency-depth target. The earlier 375/747 column mixed the two shares of the masked- A opening with logical AND/B2A openings; it was not a coherent metric. “Meaningful share bits” is the sum of protocol-defined share widths, not encoded traffic. Measured byte and direction-switch columns are separate; a prefix comparator changes correlation counts and must report its own split.

4.6 D3: Scoped public vectors and private randomness (removes G3)

The live protocol uses two distinct randomness classes. Private DPF roots, masks, and sparse noise are drawn from each process’s OpenSSL private DRBG. For the Ring-LPN public vector, a_0 is exactly the identity polynomial and is sampled and sent by neither party. Once per linear layer, each party exchanges four uniform 64-bit seed-share words; both XOR the raw shares to obtain one joint 256-bit public seed. The implemented random-oracle query is exactly

$$\begin{aligned}
Q = & \text{ASCII}(\text{RINGLPN_PUBLIC_A_V2}) \parallel 0x00 \parallel \text{seed}_{32} \parallel \text{scope_id}_{32} \\
& \parallel \text{LE64}(n) \parallel \text{LE64}(c) \parallel \text{LE64}(t) \parallel \text{LE64}(\log_domain) \parallel \text{LE64}(\text{direction}) \\
& \parallel \text{LE64}(\text{limb}) \parallel \text{LE64}(\text{slot_batch}) \parallel \text{LE64}(\text{modulus}) \parallel \text{LE64}(\text{regular}) \parallel \text{LE64}(\text{chunk}).
\end{aligned}$$

The ASCII domain plus its explicit terminal byte is 20 bytes; seed_{32} and the full scope_id_{32} are raw 32-byte values, and each remaining field is an unsigned 64-bit little-endian word in the displayed order, with $\text{regular} \in \{0,1\}$. SHAKE256 streams on Q are rejection-sampled into $\mathbb{Z}_p$ to derive the $(c-1)n$ -coefficient tail. Distinct full scope/parameter/chunk queries give independent exact-uniform vectors in the random-oracle model. For q128, both limb queries reuse the same joint seed; the encoded *limb* and *modulus* fields domain-separate them.

The short public seed cannot provide information-theoretic full-vector entropy; the real implementation therefore instantiates an explicit SHAKE256 random-oracle assumption. This removes the former multi-gigabyte public-tail exchange without silently reusing one public vector or pretending the construction is exact in the standard model. A malicious extension would replace the fixed-order semi-honest seed exchange with commit/open.

4.7 D4: Two-process deployment and endpoint/context establishment

The composed artifact runs the parties as two OS processes on distinct GPUs over two SCI-style TCP channels. Before preflight, each direction performs a versioned mutual HMAC-SHA256 challenge/response binding role, direction, invocation ID, claim digest, and fresh nonces. This rejects a rogue connector, replay, reflection, wrong role/context, or wrong shared secret at connection establishment; it does not MAC or encrypt later protocol bytes. Each process reads only its own sampled noise and emits only its own key record. The post-exit checker reads both records, reconstructs the clear masks only for validation, and invokes unchanged `readGPUMatmulKey/gpuMatmulBeaver`. The runner rejects mismatched preflight manifests, stale output, malformed or swapped records, and unilateral publication, and removes raw party records after validation.

Measured application traffic includes DPF Phase-A/B OT payloads, epoch-zero Gilboa OLE payloads, four public-seed words per party and layer, cross-term derandomization openings, conversion correlation and online messages, and final synchronization. It excludes selected-backend base-OT setup and TCP/IP overhead. Total recorded transport includes the separately counted 43,658 base-OT setup bytes. An unexecuted two-host launcher tunnels the streams over pinned SSH; current retained rows are same-host loopback and do not establish authenticated protocol-channel integrity or host isolation.

4.8 D5: Security parameter audit (G4 remains open)

The assumption behind step 1 of §3.2 is **reducible module-Ring-LPN** over each deployed prime-field limb: $(a_1, \dots, a_{c-1}, e_0 + \sum_{i=1}^{c-1} a_i e_i)$ is pseudorandom for regular sparse secrets e_i . The public vector is exactly $a = (1, a_1, \dots, a_{c-1})$; the former all-uniform- c construction is not the BCG+20/component distribution and carries no reviewed assumption here. The q128 path is two prime-field instances whose public vectors are independently domain-separated outputs from one shared joint seed in the SHAKE256 random-oracle model, followed by CRT; it is not one instance over a 124-bit field, and its limbs are not independently seeded. The splitting of $X^n + 1$ buys slot packing but exposes reductions modulo sparse low-degree factors. This is the BCG+20 reducible-ring setting, not directly the quasi-abelian group-algebra setting of Bombar et al. [23].

Parameter names matter: BCG+20's total weight is $w = ct$, so its Table-1 row $(c, w) = (4, 64)$ maps to this implementation's per-polynomial $t = 16$, not $t = 64$. The corrected full version does not provide one reproducible projection rule: Section 8.2 derives $cd(1 - (1 - 1/d)^t)$, whereas Section 9.1 uses $w - cd + (c(d - 1) + w)(1 - 1/d)^{t-1}$; its literal smallest-factor test selects degree 16 for this row, while Table 1 reports degree 128.

A source audit also invalidated several preliminary estimator outputs. The accepted EURO-CRYPT 2024 aggregate calls attack formulas containing $\binom{N-k}{t'}$ and $\binom{N-k-1}{t'}$, so a projected tuple must at least satisfy $t' \leq N - k - 1 = d - 1$. The artifact silently evaluates out-of-range combinations. Thus the recorded 57.293-bit $(c, t, d) = (4, 16, 16)$, 218.641-bit $(4, 64, 64)$, and 257.023-bit $(4, 34, 64)$ values are invalid function calls, not attack costs. Mechanically defined rows include 135.12 bits at $(4, 16, 64)$, BCG's degree-128 row at 145.85 bits, and 190.53 bits at $(2, 128, 256)$, but these remain hypothetical finite-field regular-noise model outputs rather than Ring-LPN security estimates.

The exact regular-projection law remains pinned as a finite combinatorial statement: t independently occupied source positions map into d buckets, with the corresponding occupancy distribution and independent uniform nonzero coefficients in $\mathbb{F}_p^*$ conditioned on projected support. It covers collisions and cancellation. It is freshly source-bound to live `two_party_spfss.h` SHA-256 prefix `fdbb56f8`: a semantic diff from the prior audited revision changes only optional Phase-C

OLE-source forwarding in the CPU baseline; `validate_party_noise` and `sample_party_noise` are unchanged. This establishes current distribution correspondence, not a security pin.

The source-pinned 2024 regular-ISD calculator and exact automorphism-orbit law remain model diagnostics. The self-tested 2025 hybrid-RSD formula artifact was freshly regenerated: its script/CSV SHA-256 prefixes are `cbcedaf6/1f671d94`, with the script digest embedded in every row. Its 20 rows are not an attack implementation, quantum cost, or structured-code reduction, and no reviewed newer-attack instantiation maps these diagnostics to the live tuple. Therefore D5 has not pinned n, c, t, p_0, p_1 , and this proposal makes no 128-bit classical or quantum claim. The measured $n = 2^{17}, c = 4, t = 34$ implementation NO-GO remains valid independently: regular noise requires $t \mid n$, and the uniform layout would materialize 7,272,923,136 host slots (at least 123.6 GB in one process). No replacement parameter is pinned.

4.9 Composition status

boundary	status	executable/proof evidence	consequence
G1 centralized keygen	closed for forward FC	live distributed GPU-AES DPF keys in six cases and the model run; exact correction-word coupling	no centralized key generator; trusted-dealer-free only in stated RO model
G2 share conversion	implemented; proof open	exact source algebra and corrected local inverse; full conditional transcript missing	no completed P-CONV privacy proof
G3 shared/private RNG	closed in RO hybrid	per-party OpenSSL DRBG, unsent identity $a_0 = 1$, four-word joint seed, and independently scoped SHAKE256 rejection sampling	exact required distribution in the stated random-oracle model
process/state separation	executable; peer isolation open	distinct processes, GPUs, input files, and output records, but the current loopback runner uses one UID; a fail-closed two-host isolation harness exists but has not been executed	functional composition only, not an OS isolation claim
G4 concrete parameters	open	source audit rejects prior pins; no reviewed replacement	no security-level claim
G5 graph/state and broader scope	known-zero composition closed; setup open	trusted nonlinear adapter plus the complete stock graph and terminal reconstruction	nonlinear setup and private or trained models remain open

Table 4: Current composition status. “Closed” is scoped to each row’s stated boundary: the theorem is forward-linear, and graph closure is one known-zero control. Deployment still requires distinct hosts, authenticated transport, and a reviewed Ring-LPN parameter pin.

4.10 Per-layer cost model (worked example)

For one FC layer (M, K, N) at L limbs, the current self-sustaining artifact uses

$$2L \left\lceil \frac{MKN}{C_{\text{app}}} \right\rceil$$

Ring OLEs. Application derandomization sends $4MKNL$ field words. Every Ring-OLE instance after the first per limb also consumes C_{key} earlier slots and sends two masked-difference field words per consumed slot. Conversions equal MN (each per Table 3); key-generation material per Ring OLE is in Table 2. The online phase is unchanged because keys are byte-compatible.

Worked executable example, $16 \times 32 \times 16$ at q64 ($L=1$, $\text{bw}=16$, $n=8192$, $c=2$, $t=8$): $MKN = 8192$ cross terms per direction need four Ring OLEs at $C_{\text{app}} = 7424$. They provide 29,696 application slots, use 16,384, and report 13,312 unused. Of 3,072 reserved slots, 2,304 bootstrap the three post-base instances and 768 are terminally discarded. Application derandomization sends 32,768 field words; bootstrap adds 4,608 masked-difference field words. The $MN = 256$ conversions use 31,744 AND triples plus 256 daBits. These capacities and consume/discard equalities are executable CSV gates, not a projected production epoch.

5 Experiment setup

5.1 Platform and environment

All measurements in this document were produced on one host: $4 \times$ NVIDIA RTX 5000 Ada (32,760 MiB each, compute capability 8.9; driver 560.35.03), CUDA 12.6, Linux 6.8, and a 32-core CPU allocation. Builds run on the host. Two-party tests run as separate processes pinned to distinct GPUs through `CUDA_VISIBLE_DEVICES` over single-host IPv4 loopback. The canonical q64/q128 and classifier suites use SCI/IKNP. A separate current EMP-Silent five-case/16-control rerun is opt-in and independently unreviewed. Both backends use `PartyChannel`/SCI `NetIO`; local measurements authenticate endpoint/context before preflight but do not protect later bytes at the application layer. Application-byte counters exclude backend setup and TCP/IP overhead. The GPU NTT backend is a cheddar-derived merged-stage kernel family (signed Montgomery arithmetic; q32/q64/q128-CRT; negacyclic) with the Hadamard product adaptively fused into the inverse-NTT load and forward NTTs of fixed public vectors cached across expansion iterations.

5.2 Validation methodology

Three rules, applied to every artifact. (1) *Independent oracles*: fast implementations are validated against separately written references — the GPU NTT against a host reference (`host_polymul_reference`), the generator’s internals against host suites (135/57/36 host validation cases), the end-to-end pipeline against Orca’s *unchanged* online code reconstructing the correct product. (2) *Party-separation discipline*: transcript artifacts compute each party’s view in separated state; any read across the boundary is an explicitly named oracle. (3) *Mechanical gates*: every artifact ships source + build script + run script + CSV/MD/log; suites exit non-zero on failure. The canonical checkpoint revalidates all host and GPU component gates:

```
RUN_GPU_SMOKE=1 REQUIRE_GPU_SMOKE=1 PATH=/usr/local/cuda/bin:$PATH \
scripts/run_paper_checkpoint_smoke.sh
```

and must print `ALL GATES PASS`. The exact-shape repeated evaluation is separately regenerated by `scripts/run_two_party_fc_model_scale.sh`.

Pinned reproduction boundary. `scripts/Dockerfile.reproduction` contains no source or credentials and binds the CUDA base digest, dated Ubuntu snapshot, exact package set, GPU architecture, and TeX toolchain. A clean recursive clone is mounted at runtime; `scripts/reproduce_publication.sh` rejects dirty source, missing or untracked checkpoint evidence, checksum drift, root-owned outputs, stale destinations, a nonconforming GPU, any gate mutation outside regenerable result paths, and any surviving private key, state, ledger, or scratch artifact. The shared FC/Conv binary approval binds two byte-identical builds whose command, working-directory, and build-environment receipts use only canonical clone, source, and build-root tokens; a self-consistently rehashed receipt with a raw workstation path is rejected before record execution. Its `local-smoke` mode rebuilds this PDF and the full canonical gate but ends with an explicit “not two-host publication evidence” marker. Publication mode additionally requires an owner-authorized annotated source tag, an operator-supplied tag/commit consistency object whose shape and self-digest are checked (no cryptographic signer is verified), an immutable runtime image and in-image binary identity, pinned SSH host identity, a separate owner-only persistent read-write ledger mount, and the authenticated two-host/checker manifests. Those release and deployment identities have not been issued; this remains an internal reproducibility contract rather than released two-host evidence.

5.3 Verified artifacts and matched baseline

Single-process real-generator diagnostic. The older suite² validates the Ring-LPN engine and slot/Orca algebra in nine q64/q128 uniform/regular cases. It uses centralized DPF key generation and clear exact conversion at explicitly named O1/O2 boundaries. Its narrow stage timings are diagnostic only and are not compared with the live end-to-end measurements.

case	ring OLEs	ideal-OLE equiv.	slots used	opened $\mathbb{Z}_p$ words	contract
q64, 16×32×16, bw=16	2	16,384	8,192	32,768	pass
q128, 16×16×16, bw=32	4	8,192	4,096	32,768	pass

Table 5: Representative rows of the single-process diagnostic (`orca_fc_real_ole_transcript_transcript_suite.csv`). “Ideal-OLE equiv.” is the number of per-scalar OLE calls the same layer consumed before slot packing.

Distributed key-generation protocol-logic prototype (corrected and rerun 2026-08-06). The artifact³ implements the two-party key-generation protocol of §4.3 party-separated on the host. Cross-party values flow only through counted ideal functionalities (1-of-2 string OT, Beaver bit triples, and three scalar OLEs) or explicit openings. It emits the standard existing `spfss_host::DPFKey` format, consumed by the untouched `spfss_host::dpfEvalAll`. That evaluator uses a `splitmix64` correctness PRG explicitly labelled non-cryptographic; the artifact is therefore a protocol-logic and compatibility prototype, not a secure/private implementation.

The suite (Table 6) generates 2,432 key pairs across six configurations, tree depths 4–14, and both CRT primes. Every pair full-domain-evaluates to $\beta \cdot [x=\alpha]$. In every configuration, the first two trees deterministically cover $\alpha = 0$ and $\alpha = 2^{d_{\text{DPF}}} - 2$, factors 1 and $p-1$, and resulting payloads $p-1$ and 1; the remaining trees use random nonzero payload factors. Eight centralized references pass through the same evaluator; five independent root-seed, seed-CW, left-flag-CW, right-flag-CW,

²Artifact `bench_orca_fc_real_ole_transcript`; per-case results in Table 5’s source CSV.

³`test_distributed_dpf_keygen`; per-case results in `distributed_dpf_keygen_prototype.csv`.

and final-CW corruptions fail; and six invalid point/payload encodings abort before correlation is consumed. The field `old_sign_opening_leak_control=yes` records that the omniscient regression model reconstructed the removed sign and observed a proper party-0 leaf-control-bit class containing the true point. This demonstrates the old transcript’s distinguishing power; it is not a privacy proof. Every row also requires ideal-mask-draw accounting and a duplicate-correlation-ID negative control to pass. The gate is wired into the repository’s one-command re-validation (§5.2).

prime	depth	trees	pass	string OTs	triples	OLEs	logical open	meaningful share bits
p_0	4	512	512/512	8	3	3	588	1,176
p_0	8	512	512/512	16	7	3	1,116	2,232
p_0	11	384	384/384	22	10	3	1,512	3,024
p_0	14	256	256/256	28	13	3	1,908	3,816
p_1	14	256	256/256	28	13	3	1,908	3,816
p_1	8	512	512/512	16	7	3	1,116	2,232

Table 6: Distributed DPF key-generation host results.

Source: `distributed_dpf_keygen_prototype.csv`. For tree depth d_{DPF} , per-tree counts match $2d_{\text{DPF}}$ string OTs, $d_{\text{DPF}} - 1$ bit triples, three scalar OLEs, $2(d_{\text{DPF}} - 1) + 130d_{\text{DPF}} + \lceil \log_2 p \rceil$ logical opened bits, and $4(d_{\text{DPF}} - 1) + 260d_{\text{DPF}} + 2\lceil \log_2 p \rceil$ meaningful share bits. Every row reports `transcript_accounting=pass`, `ideal_mask_draw_accounting=pass`, and `correlation_reuse_control=pass`. Timings are approximately 2.0 ms per depth-14 tree, single-threaded and including validation; they are run observations, not transport measurements. Table 6 proves functional compatibility, edge correctness, and ideal-primitive accounting, not computational privacy or 128-bit security.

Retained classifier-layer result (2026-08-10). The recorded artifact evaluates the exact inference classifier-layer shape $1 \times 512 \times 1000$ at `bw = 32` with $(n, c, t) = (8192, 2, 8)$, q128, regular noise, $C_{\text{key}} = 768$, and $C_{\text{app}} = 7424$. It executes 69 batches and 276 Ring-OLE instances. The source implements GPU-batched distributed DPF generation with GPU-AES expansion, tree-block correction-word/leaf reductions, and Ring-OLE-output Phase-C bootstrap. Each process reads only its own private noise record, runs on its own GPU, and emits only its own final key record. A 128-bit invocation identifier and persistent consume-once correlation ledger reject replay, duplicate, stale-tail, and stale-bootstrap use. After both processes exit, the checker reconstructs the records, verifies the clear product, and invokes unchanged `readGPUMatmulKey/gpuMatmulBeaver`.

Ten measured trials followed one warmup using the default SCI/IKNP backend. The environment binds current binary SHA-256 `02eaaac9....`. These same-host timing/traffic rows authenticate endpoint roles and invocation context before preflight, but later protocol bytes have no per-message integrity. The public-vector construction exchanges four 64-bit seed-share words per party and layer, then derives a fresh domain-separated SHAKE256 tail per Ring-OLE while treating $a_0 = 1$ as public without sending it. The aggregate CSV and digest-bound metadata pass execution, record, transcript, freshness, bootstrap-accounting, and unchanged-online checks; this table remains a classifier-layer measurement, not full-model execution.

Controlled CNN2/CNN3 FC matrix (2026-08-14). The source-authored workload manifest selects CNN2 FC4 ($100 \times 256 \times 128$), CNN2 FC5 ($100 \times 128 \times 10$), and CNN3 FC5 ($100 \times 64 \times 10$). All use q128/bw32, regular noise, and the feasibility tuple $(n, c, t) = (8192, 2, 8)$. They require 442, 18, and 9 Ring-LPN batches, respectively. Every layer passes one warmup and ten measured trials, for 30/30 passing measured layer trials. Both protocol parties run on distinct quiescent RTX 5000

metric	current measured value
warmup / measured trials	1 / 10
mean setup-included critical path / sample SD	4.011203588 s / 0.036601967373448 s
95% Student- <i>t</i> CI for mean	[3.985020117867291, 4.037387058132709] s
setup-included critical-path R-7 IQR	0.03481908225 s
median setup-included critical path	4.0193924415 s
median setup-included preprocessing/dealer ratio	268.6769431352700×
median shape-matched <code>gpuKeygenMatmul</code>	14.73535 ms
median unchanged two-share online	1.14969 ms
application bytes, both directions	182,372,344
total recorded transport	182,416,324
public- <i>a</i> 64-bit seed-share words, both parties	8
Phase-C products per party, epoch-zero / PCG	1,536 / 210,432
reserved slots per party, consumed / terminal discard	210,432 / 1,536
unused application slots per party	1,024
semantic protocol dependency layers	11,023
peak host / device-wide GPU-used bytes	142,542,848 / 31,929,597,952
final payload bytes per party	4,108,096
serialized record bytes per party	4,108,304

Table 7: Exact retained ResNet18 classifier-layer result. Time statistics are over ten measured trials; 268.6769431352700× is the median of setup-included shape-matched per-trial ratios, not the quotient of displayed medians. Setup-included time is the per-layer sum of the party maximum of `total.us+preflight.us+ot_setup.us`. It excludes `PartyChannel` construction, socket establishment, and channel authentication, so it is not true end-to-end time. GPU occupancy was uncontrolled and the runs were not controlled on the same physical GPU, so this is not a hardware-matched A/B comparison. The environment binds current binary `02eaaac9...`. Application bytes exclude selected-backend base-OT setup and TCP/IP overhead; total recorded transport includes 43,658 base-OT setup bytes. The GPU-used value is a device-wide observation that includes unrelated shared-GPU occupancy, not a process allocation measurement.

Ada GPUs. After they exit, the stock `gpuKeygenMatmul` comparator runs on the quiescent physical GPU of the sample’s slower setup-included party; advisory per-GPU locks span the complete run. Thus the comparator matches source shape, key contract, binary, allocator and synchronization boundary, physical GPU, and observed quiescence. Execution order is fixed as protocol then dealer, not randomized.

The original August raw CSV and its schema remain unchanged. The corrected observation, summary, output-schema, and reconstructed analysis-plan files have the stem `results/fc/two_party_fc_model_scale_cnn2_cnn3_` and suffix `2026_09_11`; the historical source-manifest inputs are retained in `results/fc/model_scale_inputs_2026_08_14/`. The live aggregator emits layer-invocation observations (v4) and independent-layer statistics (v3). Reversing only one layer’s trial labels leaves every estimate and interval unchanged. The previous CNN2 synthetic model-trial distribution and paired-ratio median are withdrawn, not pooled with this analysis. Per-layer measured distributions remain meaningful.

In the retained classifier rows, legacy post-OT median Phase C is 0.041723s, Phase B is 1.955515s, Phase A is 0.2462935s, and GPU Ring-LPN expansion is 1.226650s.

Full-graph systems control. One retained source-bound q128/bw32 known-zero run consumes a fresh 21-record ResNet18 convolution/FC set through the unchanged Orca linear consumers. Its

metric	CNN2 all FC	CNN3 FC5
FC layers / measured layer trials	2 / 20	1 / 10
sum of layer mean setup-included times (s)	26.881842	0.903894
95% stratified-bootstrap CI (s)	[26.849728, 26.911767]	[0.892967, 0.915280]
sum of layer mean stock-dealer times (ms)	30.46697	14.92146
setup/dealer ratio of summed layer means	882.3274×	60.5768×
95% stratified-bootstrap ratio CI	[876.3872, 890.1396]	[59.3419, 61.8764]
application / total transport bytes	1,302,752,736 / 1,302,840,696	39,432,504 / 39,476,484
semantic dependency layers	72,278	1,663

Table 8: Controlled FC feasibility matrix, reanalyzed September 11. Layers were measured separately; integer trial labels do not form joint model trials. Model estimates sum independent layer means, not model-trial latencies. Intervals use 10,000 whole-row resamples within each layer (seed 20260911), preserving each protocol/dealer pair; endpoints use R-7 percentiles. No model median, SD, IQR, or Student-*t* interval is inferred. The per-layer critical-party timer is `total_us+preflight_us+ot_setup_us`; it still excludes channel construction, socket establishment, authentication, checker time, and the stock dealer. “Semantic dependency layers” are implementation schedule depths, not packet or network-round counts. Total transport adds selected SCI/IKNP base-OT setup bytes but excludes TCP/IP framing. The fixed protocol-then-dealer order can retain order bias. All values are local loopback feasibility measurements at unpinned parameters, not full-model, WAN, security-level, or competitive-performance evidence.

execution manifest overrides the source tuple from $(n, c, t) = (8192, 2, 8)$ to $(262144, 2, 8)$ so the largest layers fit; this is an execution parameter, not a security pin. The compiled graph contract is generated from, and hash-checked against, the same source manifest used by the record-set gate. Its explicit value-source registry binds every main and shortcut operand; the exact stock forward order contains 21 linear calls, 21 stochastic truncations, 19 stock key consumers (one MaxPool, 17 ReLUs, and classifier sign extension), 20 remask edges, eight residual merges, integer GlobalAvgPool2D, and terminal reconstruction.

The 21-record set contains 6,439 Ring-LPN batches, 25,756 Ring-OLE instances, 6,593,536 DPF trees, and 130,774,336 final payload bytes per party. In this one sequential run, the sum of legacy post-OT linear-layer critical paths was 4,720.417 s. The 50.161-s full-graph orchestrator wall measurement covers the graph positive run and seven fail-closed graph-output controls; it excludes record generation and is not positive-path end-to-end latency. The record-consuming graph critical path was 11.852 s. These are not a timing distribution or competitive result.

A test-only trusted compatibility adapter reads both parties’ source-bound linear mask states, supplies both parties’ truncation successor-mask shares plus remask and terminal material, and emits the exact 62-item stock key stream, including all nonlinear keys. It took 9.356 s and produced 1,084,582,352 raw stock-key bytes per party. Each live party reads only its own private linear, nonlinear, and persistent-state records. A separate post-exit checker independently reconstructs every linear input/output, remask, residual, pool, sign-extension, and terminal value from both published records and rejects counter/order drift. Seven fail-closed graph-output controls—forced second rename, reused invocation, stale output, party-record swap, truncated nonlinear record, nonlinear payload corruption, and digest-valid trace corruption—all reject without partial output; source-manifest and topology mismatch controls also reject.

The retained 17-indexed-payload-file bundle plus `INDEX.json` at `results/graph/resnet18_full_graph_checkpoint_2026_08_10/` excludes private linear, mask-state, nonlinear, ledger, and output records. Its original manifest-bound adapter provenance retains workstation/build-root

strings, and the historical linear assignment reused GPU 1 sequentially for party 1 and its checker. These are non-secret internal/advisor metadata, but they block external circulation and no longer satisfy the current retained-assignment policy. The original bytes are preserved for transitive manifest/approval/INDEX integrity; a fresh canonical three-GPU run with normalized provenance must replace this checkpoint before release. The orchestration deliberately applied each omniscient per-record correctness checker before feeding the same test record to the graph control, so it does not evidence consume-once online deployment. The trusted nonlinear key source means this is not dealerless nonlinear preprocessing or added security evidence; known-zero inputs and weights mean it is not private/trained inference, an accuracy result, deployment evidence, or a full-model performance claim.

The closest in-artifact dealer comparator is stock `gpuKeygenMatmul`, aligned on shape, masks, bit width, allocator/synchronization boundary, and output bytes. The retained classifier execution was not controlled on the same physical GPU and occupancy state, so its $268.6769431352700\times$ statistic remains descriptive. The newer CNN2/CNN3 matrix adds per-sample physical-GPU and quiescence control; its $882.3274\times$ and $60.5768\times$ ratios of summed layer means are controlled, strongly negative stock-dealer comparisons, not speedup claims. A separately pinned stock libOTe Reverse-Cuckoo run and a project-prime exact- p_0 native-folded adapter diagnostic are retained:

row	setup	online/full domain	protocol total
stock Goldilocks		12.43 s process wall	
exact p_0 native-folded	18.523424 s	2.116894 s	20.640318 s

Table 9: Reverse-Cuckoo baselines at nominal $(n, c, t) = (2^{20}, 4, 16)$. The stock row uses internally sampled factors, synthetic preloaded base correlations, CPU local sockets, Goldilocks arithmetic, and a 16-folded layout. The exact- p_0 diagnostic uses 256 sets of 16 points (4,096 terms), checks 16,777,216 full-domain positions, and records 52,791,184 setup plus 1,425,584 online wire bytes. Its protocol total is 20.640318 s and its validation-inclusive total is 20.688314 s. Neither row is raw 31-diagonal GPU output, a function-matched comparison, or a security-level result; no speedup ratio is reported.

Optional EMP-Silent feasibility rerun (2026-08-14). The current opt-in rerun binds FC binary `bf4e4f90...`, authorized bridge `435f3be6...`, and EMP-OT revision `2fca139f...`. All five live cases and all sixteen rejection controls pass; every declared directional 128-bit-OT inventory is consumed exactly. There is one trial per case and all four GPUs carried unrelated resident work-

case	straight OTs	reversed OTs	P_0 sent	P_1 sent
q64 regular small	59,120	11,248	1,716,463	550,112
q64 uniform small	62,192	14,320	1,766,383	600,032
q128 regular small	118,232	22,496	2,645,481	739,129
q128 uniform small	124,376	28,640	2,745,321	838,969
q64 regular multibatch	93,184	37,376	2,159,029	739,717

Table 10: Focused EMP-Silent accounting. Byte columns sum instrumented backend correlation, chosen-message adjustment, and ciphertext bytes sent; they exclude channel authentication and TCP framing. The selected API exposes no nonoverlapping base-OT subboundary. Inventories are equal at both parties.

loads. Timing/memory columns therefore support no comparison. The custom SilentFerret revision

remains independently unreviewed, so these rows make no performance, bandwidth-improvement, security, or headline claim.

Performance anchors. Ring-OLE expansion ($n=8192$, $c=2$): 9.086 ms (q64) and 18.230 ms (q128) for regular noise at the $t=8$ smoke point, one iteration per configuration ($n=1$ each); separate $t=64$ diagnostics record 881 ms uniform and 61 ms regular, with two timed iterations per mode ($n=2$ each; four total), and pair-key bytes 9,044,160 (9.04 MB) and 5,529,408 (5.53 MB), respectively. The distributions and SPFSS domains differ, so no ratio is claimed. SPFSS full-domain evaluation dominates ($\sim 99\%$); the NTT is under 1% — measured directly, and confirmed by the fact that a 2–8% NTT-side improvement left expansion time unchanged. The oracle-backed conversion stress artifact is bit-exact on 8,516 conversions per configuration; the live two-process OT-backed artifact passes 76 conversions in each of four q64/q128/bit-width configurations, at the costs of Table 3. Byte-compatibility: with the integration flag off, baseline Orca output is byte-identical (verified through the real two-party FC test).

5.4 Status gates and remaining experiments

Closed — current live forward path. The current focused two-process SCI/IKNP suite passes all five q64/q128 cases and controls, generating distributed GPU-AES DPF keys, self-bootstrapped Ring-LPN OLE shares, converted q32 masks, and byte-compatible Orca FC records. Every focused row records positive breadth-first call counts for both parties and zero root-to-leaf calls; all 21 shape plans pass. The current FC/Conv binaries begin `02eaaac9/975ac726`, and the focused approval digest begins `ce3cc3a5`. These are correctness/path-counter results, not current Conv0 timing or a breadth-first speedup claim. Sixteen negative controls cover endpoint/context authentication, replay, reflection, freshness, ledger integrity, nonpositive capacity, malformed/swapped records, and publication failure; the unchanged online consumer reconstructs the expected product.

Closed narrowly — model-FC matrix. The source manifest selects CNN2 FC4 and FC5 and CNN3 FC5; each passes one warmup and ten measured trials. Per-sample stock-dealer execution follows both parties on the quiescent physical GPU of the slower setup-included party. Ratios of summed layer mean times are $882.3274\times$ for CNN2 and $60.5768\times$ for CNN3. This closes source coverage, per-layer repetition and dispersion, and same-hardware/quiescence control for these three FC layers, not competitive performance, parameter security, full-model execution, or network evaluation.

Closed narrowly — full-graph systems control. One source-bound known-zero run binds a fresh 21-record set to all unchanged Conv2D/FC consumers, all secure truncations and stock nonlinear consumers, every projection/identity residual merge, GlobalAvgPool2D, sign extension, and terminal reconstruction. Manifest/topology, publication, nonlinear-record, and output-corruption controls reject. The trusted test-only adapter reads both mask states, supplies both truncation successor-mask shares plus remask/terminal material, and generates the nonlinear keys. Therefore dealerless nonlinear preprocessing remains open.

Closed conditionally — proof composition. The theorem composes standard semi-honest DPF, OT/OLE, conversion, DRBG, and authenticated-channel assumptions with a selected Ring-LPN sample distribution. The exact regular-projection law is freshly bound to the current sampler, and the self-tested hybrid-RSD formula evidence is freshly regenerated. The

structured-code hardness transfer, concrete parameter pin, and independent cryptographic review remain open.

Open — parameters. A reviewed structured-code and two-CRT-limb advantage analysis must pin (n, c, t, p_0, p_1) before any security-level claim. All performance rows must then be rerun because the current point is feasibility-only.

Open — performance engineering. Phase B remains the largest measured stage, followed by GPU Ring-LPN expansion and Phase A; Phase C is much smaller. Full-duplex Phase-A/B scheduling and retained asynchronous frontier allocation are complete. A source- and cost-audited specialized regular-DMPF design attempt returned NO-GO: no reviewed candidate preserved the fixed transcript, plain Ring-LPN boundary, application-slot budget, stock-key ABI, and deployed-shape cost simultaneously. Keep the measured DPF path for the approved full-linear systems integration and pursue algorithmic Phase-B/round reduction. The controlled FC matrix reports setup-included preprocessing/dealer ratios of summed per-layer means of $882.3274\times$ for CNN2 and $60.5768\times$ for CNN3. The result is not competitive; reruns at reviewed parameters and further Phase-B/dependency-layer work remain necessary.

Open — model realism and dealer scope. Exact source-bound ResNet18 graph coverage is closed for one known-zero input/weight control. It is not private or trained inference, accuracy evidence, a timing distribution, or full-model performance. Dealerless nonlinear key generation remains absent; repeated clean runs with private inputs and trained weights, including an accuracy-bearing evaluation, are still required before a broader systems claim.

Open — deployment. Source contains fail-closed two-host coordination and peer-private container orchestration: both complete post-handshake streams are carried through one pinned-host-key, AEAD-only OpenSSH tunnel with bounded rekeying. No configured second host was available to execute it. Current evidence uses same-host loopback with authenticated endpoint/context establishment but no application-layer protection for later bytes; it cannot predict LAN/WAN latency, establish executed authenticated-channel integrity, or prove OS-level peer isolation.

Open — review. A human cryptographer who did not author the artifact must review the DPF/OT/OLE/composition proof, the structured-code parameter argument, and the custom pinned EMP-Silent revision. Internal adversarial review and focused differential/correctness tests are not substitutes.

Out of scope. Training-state updates, a dealerless DCF-key protocol, active security, and full-model dealer removal require separate designs and evaluations.

5.5 Reporting and claims discipline

The current supported statement is: “At unpinned feasibility parameters, a live same-host two-process implementation generates party-local forward-linear keys without a trusted dealer in the stated random-oracle model; those keys match the stock Orca format and pass unchanged online computation. Endpoint roles and invocation context are mutually HMAC-authenticated before preflight, but later protocol bytes have no per-message integrity. A separate source-bound known-zero control composes a fresh 21-record linear set with the exact stock ResNet18 graph using a trusted test-only adapter that owns both truncation successor-mask shares, remask/terminal material, and nonlinear keys. Conditional semi-honest security applies only to the forward-linear

theorem under its explicit DPF, OT/OLE, authenticated-channel, and exact-distribution Ring-LPN assumptions.”

The in-artifact quantitative dealer comparator is stock `gpuKeygenMatmul`. The current CNN2/CNN3 matrix aligns source-authored shapes, bit width, masks, output bytes, binary, allocator/synchronization boundary, physical GPU, and observed quiescence for every sample. Its fixed protocol-then-dealer order is not randomized, and its strongly negative ratios remain local feasibility results. The older classifier rows lack physical-GPU and occupancy control and remain descriptive only. The older single-process Ring-OLE stage timers omit most live work and are not a baseline. Prior dealerless PCG systems expose different correlations, parameter assumptions, and applications; no compatible implementation has been measured under this artifact’s contract, so no cross-paper speedup ratio is reported.

Unsupported wording includes “128-bit secure,” “faster than the dealer,” “WAN-ready,” “dealerless full Orca,” and any dealerless-nonlinear, nonlinear-security, training-layer, private-inference, accuracy, or full-model performance/security claim. Every future performance table must identify topology, trust/oracle boundary, coverage, exact parameters, counter scope, warmups/trials, and environment; q64/q128 denote one/two 62-bit arithmetic limbs, not security.

6 Conference-readiness contract: research hypothesis and evidence gates

Verdict and venue rules. This remains **not submission-ready**; the following is an internal research contract, not a prediction of acceptance. ASPLOS 2027 requires substantive advancement of architecture, operating systems, or programming languages, not merely their use; its rapid review reads only the first two pages [27]. A future submission must therefore make the systems question, new mechanism or insight, strongest evidence, and trust boundary self-contained there. The CFP specifies an 11-page main paper using `acmart` with `sigplan,anonymous,review,nonacm`, no template squeezing, and a September 9, 2026 AoE submission deadline. That deadline has passed as of this September 11 supplement. Appendices cannot carry essential evidence omitted from the limit, and reviewers need not read them. This historical internal article retains its format, date, author, and circulation restriction; recording these rules neither converts it into a submission nor grants permission to circulate it.

Candidate accelerator-systems question—a hypothesis. Can a GPU execution schedule for distributed correlation generation turn available tree-level parallelism into useful *application-ready* offline throughput once protocol dependencies, host/device movement, finite memory, freshness, and secure transport are charged? The hypothesis is that the boundary between GPU compute throughput and communication/dependency cost predicts when batching and retained execution state help, and when additional GPU parallelism cannot improve the critical path. Neither this predictive boundary nor a novel scheduling contribution is established here. The retained live profile identifies Phase B as largest, followed by GPU expansion and Phase A, with Phase C much smaller (§5.4); the older isolated-expansion profile is not a profile of the composed protocol. These observations motivate causal experiments, not an architectural result.

Executed scheduling experiment—September 11 reproducible supplement. We replaced per-duplex-call SCI sender-thread creation with one lazy, channel-owned sender worker. The caller still waits for both directions; the worker borrows a stack callable only until completion and introduces neither a per-job allocation nor a payload copy. This is a host scheduling

optimization, not a new cryptographic protocol or an established GPU architecture contribution. On one shared RTX 5000 Ada host, the CNN3 FC5-shaped $100 \times 64 \times 10$ workload at q128/bw32 and $(n, c, t) = (8192, 2, 8)$ passed one warmup and ten measured invocations per version. Five measured pairs ran before–after and five after–before. Every invocation used fresh private state; the independent stock-consumer checker ran only after both parties exited. All 20 measured checks passed, with identical values for 68 per-party protocol/contract/accounting fields, including OT inventories, bytes, dependency counts, and evaluator counts.

Median per-invocation metric	Before	Reusable worker
Process launch through both party exits	0.976022 s	0.906485 s
Post-channel setup-included preprocessing	0.858949 s	0.790864 s
Slower-party DPF Phase B	280.205 ms	223.252 ms

The ratio of process-latency medians is $1.07671\times$ (7.12% lower latency); the median of the ten *paired* ratios is $1.07838\times$. Process-latency sample standard deviations are 10.02 ms before and 11.49 ms after. Phase B’s median decreases by 20.33%. No confidence interval from the previous experiment is reused. These timers have different scopes and are not summed. The retained `scripts/run_fc_review_experiments.py` controller times from immediately before the first producer launch through blocking observation of both timeout-wrapped producer exits. Private-fixture creation, binary hashes, CSV parsing, the checker, and GPU-idle admission are outside that timer. Raw warmup/measured samples and the pre-execution frozen plan are `results/fc/sci_duplex_worker_reproducible_2026_09_11.json` and its `.plan.json` companion. They bind the harness, binary and source identities, calculation rules, timer boundaries, and all 22 invocations. Idle admission is not an exclusive GPU lease, fixed-clock, or CPU-affinity control; the host remained shared. The September 10 JSON remains historical: its controller source was not retained, and its measurements are not pooled with this reproducible experiment. This one-shape result supports the scheduling change locally, not a matched-security dealerless comparison, a full-model speedup, a clean-clone release attestation, or a general resource/dependency model.

Prospective capacity prediction—executed and rejected. Before any new measurement, the retained controller froze eight training FC shapes and two unseen held-out shapes, q128/bw32 regular $(n, c, t) = (8192, 2, 8)$, one warmup and ten measured invocations per shape, fixed ordering, binary/harness/source identities, fitting rules, and failure criteria. This is a *fixed-tuple wall-time/capacity model*, not a GPU resource or causal attribution model: FC rows do not expose GPU work and host/device-transfer counters for that stronger claim. For an $M \times K \times N$ FC shape, let

$$X = MKN, \quad C = MN, \quad W = MK + KN + MN, \quad R = 4 \lceil X/7424 \rceil.$$

The predictor is $\hat{Y} = \beta_0 + \beta_R R + \beta_X X + \beta_C C + \beta_W W$, with nonnegative coefficients fitted to training-shape medians of $Y = \max_b(\text{total.us}_b + \text{preflight.us}_b + \text{ot.setup.us}_b)$. The controller enumerates the five-column NNLS faces using column-scaled, reorthogonalized QR and checks rank and KKT conditions; no ridge fallback is used. Training covers CNN3 FC5, ModelB FC5, CNN2 FC5, P-SecureML FC3, AlexNet gemm17, P-LeNet FC5, ModelB FC4, and CNN2 FC4; exact dimensions and constructor/batch-source anchors are in the frozen plan. After all 88 training invocations, but before either held-out warmup, it exclusively wrote and fsynched the training seal, coefficients, and pointwise 95% prediction intervals. The intervals use 10,000 centered training-median residual resamples (seed 20260911), NNLS refitting, an independent residual draw per held-out prediction, and R-7 endpoints. They assume exchangeable shape-median errors, not simultaneous coverage or calibrated within-shape or heteroskedastic uncertainty.

Held-out shape	Prediction (s)	Median (s)	Sealed interval (s)
P-SecureML $128 \times 128 \times 128$	14.57796	14.35828	[14.21224, 15.28882]
AlexNet $100 \times 256 \times 256$	42.93165	43.02620	[42.88603, 42.97896]

All 110 producer/checker invocations and accounting checks pass; the **prediction gate fails**. Point errors are 1.530% and 0.220%, below the frozen 15% threshold, but AlexNet’s measured median lies outside its sealed interval. Close point predictions do not rescue the rejected interval contract. No samples were excluded, held-outs reclassified, thresholds widened, or parameters retuned. The interval procedure is not validated by this result. Raw samples, the pre-execution plan, and the training seal are `results/fc/capacity_model_prospective_2026_09_11.json` and its `.plan.json/.seal.json` companions. The seal and every training-sample hash replay exactly from the retained data. Parties used GPUs 1/2 and the post-exit stock checker GPU 3; shared-host/no-fixed-clock limitations apply. This negative experiment does not establish cross-hardware prediction, WAN behavior, a reviewed-security baseline, or an ASPLOS contribution.

Claims, present evidence, and decisive missing experiments.

Compatibility, narrowly supported. The unchanged linear consumers and known-zero graph control establish the boundaries in §5.5. A broader functional claim needs repeated nonzero private-share executions with a separate fixed-point reference, checking signed values, wraparound, truncation, residual joins, and terminal output at the claimed boundary. Oracle reconstruction belongs only in post-run validation, not the deployed path. Trusted nonlinear/truncation material must remain explicitly labeled; private trained-model accuracy or dealerless full inference requires its own evidence and, for the latter, removal of that trusted source.

GPU benefit, unestablished. Compare the same cryptographic protocol and output contract on GPU and an optimized cryptographic host implementation; the splitmix64 logic prototype is not such a baseline. Cross GPU/host execution with epoch-zero and amortized operation, holding parameters, useful output count, security, transport, and correctness checks fixed. Isolate scheduling, retained allocation, and batch size while measuring stage critical paths, GPU work, host/device transfer, synchronization, peak memory, and bytes. Do not sum overlapping stage timers as wall time. Explain both wins and losses, then test the resulting cost model on held-out shapes and an additional hardware configuration before claiming generality.

Dealer-removal cost, partly measured. The controlled CNN2/CNN3 stock-dealer results in §5.3 are negative local feasibility evidence. Stock `gpuKeygenMatmul` is a *trusted-dealer lower-bound reference* for removal cost, not a matched-security competitor or a proven mathematical lower bound. A dealerless comparison must match concrete security target, adversary and channel assumptions, arithmetic/correlation semantics, usable output, and setup scope; include any conversion, packing, serialization, or authentication needed to reach the same consumer. Pin implementation revisions and disclose irreducible assumption differences. The retained Reverse-Cuckoo diagnostics do not meet this contract; until an executable matched baseline exists, no competitive dealerless ratio is supported.

Accounting and network experiment contract (proposed project gates). Report application-ready records or useful correlations per second, not raw expanded slots alone; show

reserved bootstrap slots, discarded tails, and capacity utilization. For K fresh invocations, report both epoch-zero latency and $(S + \sum_{i=1}^K P_i)/K$, where S is only demonstrably reusable setup and P_i includes all per-invocation preprocessing. Show the measured range of K , rather than extrapolating infinite amortization. Charge initialization, base OT, distributed key generation, bootstrap, expansion, conversion, serialization, and transport exactly once at their actual lifetimes; separate one-time setup from recurring work without hiding either. Report unchanged online latency separately and total setup/offline/online service cost. Online amortization may reuse only state authorized by the proof, never consume-once keys or correlations. Any claimed offline/online overlap needs an executed pipeline, measured buffering and memory, and sustained producer and consumer rates; faster kernels alone do not establish service throughput.

Execute both complete protocol streams across distinct hosts with authenticated integrity after the handshake, pinned peer identities, and peer-private state; retain topology/configuration receipts and rejection evidence for wrong peers and interrupted/corrupted transport. Source-only SSH orchestration and local endpoint authentication do not pass this gate. Measure a declared bandwidth and RTT grid on that deployment, including an unshaped LAN point and controlled latency/bandwidth constraints, and label emulation rather than calling it a real WAN. Record achieved RTT/bandwidth, wire and application bytes, timing distributions, and critical-path changes. Semantic dependency counts are not measured rounds and cannot alone predict WAN time. Use justified workloads, repeated fresh trials, randomized or counterbalanced comparator order, dispersion, and disclosed exclusions, following SIGPLAN’s guidance on scoped claims, fair baselines, relevant metrics, and sensitivity [30].

Scientific decision and release gates (proposed, not venue rules). An experience or negative-result paper is a possible route under the CFP, but needs *generalizable new insight*: for example, a causally validated resource/dependency model that predicts a previously unexplained failure regime across workloads. Honest slowdowns, integration effort, or a list of bottlenecks alone are not that contribution. Before claiming such a result, state its falsifiable prediction and domain, test it outside the cases used to develop it, report counterexamples, and distinguish inherited techniques from the new understanding.

1. **Security and review:** pin the exact sampler, (n, c, t, p_0, p_1) , attack models/cost conventions, sample and invocation budgets, and composed advantage including both CRT limbs. Resolve the structured-code hardness transfer and authenticate the executed full stream. A qualified human cryptographer not responsible for this implementation must document review of the reduction, parameter argument, DPF/OT/OLE composition, and selected backend revision, with material findings resolved. Rerun claimed performance at the reviewed point. q64/q128 and internal checks cannot substitute.
2. **Functional and empirical scope:** pass the nonzero boundary, matched-baseline, ablation, accounting, and network contracts above for each corresponding claim; unsupported broader claims remain excluded. A negative-result route does not require a speedup, but does require the generalizable insight and evidence just specified.
3. **Reproducibility and release:** execute the final source revision from a clean clone with pinned dependencies and fresh private state; retain source/build/evidence identities and regenerate every claimed table from raw samples. Provide a single basic-test command, expected outputs, per-result reproduction commands, resource/runtime requirements, licenses, and justified variability bounds, as described in the official artifact appendix guide [29]. Before anonymous paper review, audit the PDF metadata, links, source/package metadata, logs,

and paths for identity leaks and remove secrets without rewriting historical evidence. Prepare a separately identified, permission-cleared release. Public archival deposit with a persistent DOI is our eventual release gate; Git hosting alone does not qualify for the venue’s “artifacts available” badge. Formal AE is voluntary, follows paper acceptance, and does not determine it; its camera-ready appendix is at most two pages [28]. Anonymous submission access and later public archival access are distinct stages.

4. **Human authorization and AI disclosure:** obtain explicit human author and professor/owner sign-off on correctness, claims, contribution credit, private-project reuse, licenses, final authorship, and external circulation. Maintain a factual account of generative-AI assistance to text, code, analysis, figures, and citations for human verification; fully disclose actual use under the CFP/ACM policy, without listing an AI as an author. If placed in acknowledgments, the CFP requires that disclosure immediately before references [27]. This section is not the final disclosure or the unresolved owner ruling.

7 Related work

Orca [1] defines the host system: FSS-based 2PC ML on GPUs with dealer preprocessing. Beaver’s technique [2] underlies its linear-layer keys. BCG+20 [6] already gives the Ring-LPN OLE/triple generator, fully split slot packing, and a semi-honest distributed setup blueprint using generic 2PC plus the Doerner–shelat distributed DPF [18]. Rieder’s Silentium [7] implements an actively secure Ring-LPN Beaver-triple PCG. The MIT-licensed libOTe `dmpf` branch at commit `edb5d328...` includes `RingLpnTriple` and Reverse Cuckoo. Rivinius et al.’s Convolutions in Overdrive [8] already implements maliciously secure offline convolution triples in a different MP-SPDZ setting. These are inherited or competing primitives, not this paper’s inventions; no broad first-convolution-preprocessing claim is made. Programmable DPFs [9] reduce distributed generation to constant rounds for feasible domains. Improved DMPFs at IEEE S&P 2025 [10] optimize the multi-point case and its PCG applications. Stationary Syndrome Decoding [11] instead amortizes a shared noise support across correlated instances and explicitly targets the high setup cost of Ring-LPN Beaver generation under a different assumption. More directly, Agarwal–Raghuraman–Rindal’s fully distributed DMPF [12] targets Ring-LPN/Stationary-LPN PCGs with a semi-honest proof and prototype. A pinned libOTe stock build/run is the closest distributed implementation baseline, but its field, factor inputs, folded layout, synthetic setup, CPU transport, and output contract do not match this artifact, so no ratio is meaningful. SLAMP-FSS [13] is a second 2026 multi-point design using tree PRGs and linear systems. The current tree-per-point DPF, including its corrected three-product Phase C, is a local compatibility bug fix and GPU-batched baseline rather than an independent protocol contribution. The canonical audit leaves the professor/contributor credit and private-project-overlap ruling explicitly open; nothing here prejudices ownership, permission, acknowledgement, or disclosure. Accordingly, this paper claims neither the first Ring-LPN Beaver-triple implementation nor the first fully distributed or dealerless DMPF.

Ferret [19] and Silver [20] are candidate silent OT/VOLE transports. Mixed arithmetic–binary conversion uses `edaBits/daBits` [21] and Rabbit-style comparison [22]; PCGs directly over $\mathbb{Z}_{2^k}$ [14] may remove that conversion. Any-finite-field PCGs [15] and the fully implemented QA-SD/Walsh–Hadamard design [16] are current prime-field architectural baselines that avoid Ring-LPN’s FFT choices. Liu et al. [17] provide the field-size-sensitive and regular-noise attack estimators used in the preliminary D5 audit. Bombar et al. [23] analyze a related quasi-abelian-decoding foundation, but their group-algebra theorem is not substituted for the reducible $X^n + 1$ assumption here.

The active polynomial backend is substantially derived from Cheddar [24]; this audit retains its MIT notice and records the reconstructed source pin and local delta. GPU-NTT’s Merge and four-step algorithms are separate work by Özcan, Javeed, and Savaş [25, 26] and are currently an external measured baseline. Neither backend is a contribution of this paper.

8 Conclusion and publication assessment

The concrete result is narrower and stronger than the earlier proposal language: a trusted-dealer-free live preprocessing source path in the stated random-oracle model now spans GPU-batched distributed DPF generation, executable Ring-LPN-output bootstrap, tree-block GPU reduction, slot extraction, secure q128-to-q32 conversion, byte-compatible key serialization, and unchanged Orca forward-FC/Conv2D consumption. The current five-case focused q64/q128 suite integrates the breadth-first path with positive per-party breadth calls, zero root-to-leaf calls, and all 21 shape plans passing. The current 02eaaac9... ResNet18 classifier-layer artifact separately passes one warmup plus ten measured trials. A controlled source-manifest matrix also covers CNN2 FC4/FC5 and CNN3 FC5 with 30/30 measured layer trials passing and a same-physical-GPU, quiescence-checked stock comparator. A source-bound known-zero control composes all 21 fresh linear records with all 21 secure truncations, the exact 62-item stock stream, every residual merge, integer GlobalAvgPool2D, sign extension, and terminal reconstruction. Its trusted test-only adapter reads both mask states, supplies both truncation successor-mask shares and remask/terminal material, and generates the nonlinear keys; consequently the run proves complete graph/state compatibility, not dealerless nonlinear preprocessing or a wider security theorem. No private/trained inference, accuracy, timing distribution, or full-model performance result is claimed. A mutual HMAC preflight authenticates local endpoints/context but not later local protocol bytes. The distinct-host launcher now provides an AEAD-only pinned-SSH tunnel for both complete streams, but no two-host execution has occurred. A separate current EMP-Silent five-case/16-control rerun closes only focused correctness and exact inventory/component-byte accounting; it remains opt-in, independently unreviewed, and absent from headline rows.

This is an internal/advisor checkpoint, not a submission-ready security or systems paper. The known-zero graph/state-composition seam and the narrow CNN2/CNN3 same-hardware stock-dealer comparison are closed. The latter is strongly negative: setup-included ratios of summed layer means are $882.3274\times$ for CNN2’s two FC layers and $60.5768\times$ for CNN3 FC5. The prospective capacity-model interval gate also fails. Remaining systems gates include dealerless nonlinear setup, repeated private-input/trained-model evidence, authenticated distinct-host execution, a compatible dealerless baseline, competitive performance, and independent review. No reviewed concrete-security parameter point exists. Novelty is centered on integration/engineering of known DPF/Ring-LPN/conversion primitives. A cryptography-paper claim would require a new protocol, reduction, or parameter result. Authorship, contributor credit, private-project reuse permission, acknowledgements, and disclosure remain unresolved external gates before any circulation.

The highest-value next evidence is: (1) an independently reviewed reduction and concrete parameter search for the exact regular-projection distribution and structured projected code; (2) replace the trusted nonlinear adapter with a dealerless protocol, then repeat the exact source-bound graph under real private model/input shares, multiple clean invocations, and an accuracy-bearing evaluation before making a dealerless-model claim; (3) further algorithmic Phase-B and dependency-layer reduction with renewed stage, byte, memory, source, and proof measurements, plus independent review of EMP-Silent before selecting it; (4) authenticated two-host LAN and controlled-WAN execution under distinct peer-isolation identities; and (5) repeat the controlled

stock-dealer comparison after a parameter pin and add separately labeled closest reproducible dealerless-PCG baselines. The specialized regular-DMPF route remains a design NO-GO unless a fixed-transcript construction later meets the documented simulator, application-capacity, stock-key, and cost gates. Training, dealerless nonlinear preprocessing, and malicious security should remain explicit follow-on work unless separately implemented and evaluated.

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